

Toward Transportation Digital Twin Systems for Traffic Safety and Mobility: A Review

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Abstract—Digital twin (DT) systems aim to create virtual replicas of physical objects that are updated in real-time with their physical counterparts and evolve alongside the physical assets throughout their lifecycle. DT technology promises to significantly enhance the capabilities of intelligent transportation systems (ITSs) through improving safety and mobility. The objectives of this review article are to: 1) conduct an in-depth review of DT technology applications in the transportation domain, i.e., safety and mobility applications; 2) develop a reference architecture for a transportation DT (TDT) system based on a hierarchical concept to guide in developing TDT systems at any scale; 3) identify research challenges for each component, i.e., the physical space, the communication gateway and the digital space of the TDT system, to realize safe and operationally efficient ITS; and 4) provide future research perspectives to develop and deploy networkwide transportation DT systems considering the scale and dynamic nature of future connected and automated transportation systems. Thus, this article focuses on providing a comprehensive understanding of the requirements, reference architecture, challenges, and future research opportunities for a TDT system, starting from individual transportation assets and building up to the entire networkwide TDT.

Index Terms—Automation, connected transportation system, digital twin (DT), mobility, safety.

LIST OF ABBREVIATIONS

DT	Digital twin.
TDT	Transportation digital twin.
CAV	Connected and automated vehicle.
IoT	Internet of Things.
CPS	Cyber-Physical System.
MAS	Multiagent simulation.
ARC-IT	Architecture reference for cooperative and intelligent transportation.
DTP	Digital twin prototype.
DTI	Digital twin instance.
SDT	Supervisory digital twin.
IDT	Interactive digital twin.
PDT	Predictive digital twin

ORUs	Other road users.
GPS	Global positioning system.
P2V	Physical to virtual.
P2P	Physical to physical.
V2V	Virtual to virtual.
LoRa	Long range radio.
WPAN	Wireless personal area network.
LPWAN	Low-power wide area network.
NB-IoT	Narrow Band Internet of Things.
Wi-Fi	Wireless fidelity.
MQTT	Message queue telemetry transport.
CoAP	Constrained application protocol.
ADAS	Advanced driver assistance system.
ATSC	Adaptive traffic signal controller.
LSTM	Long short term memory.
EV	Electric vehicle.
IoV	Internet of Vehicles.
DQS	Distributed quantum sensing.
SUMO	Simulation of urban mobility.
MATLAB	Matrix laboratory.
IRS	Intelligent reflective surfaces.
PIML	Physics informed machine learning.
GAN	Generative adversarial network.
ANN	Artificial neural network.
HetNet	Heterogenous wireless networking.

I. INTRODUCTION

A. Background

THE INCREASING trend toward connectivity and automation in the realm of transportation has spurred interest in the adoption of technologies for improving the management of traffic. The advent of connectivity has enabled the collection and aggregation of data from transportation systems at all levels, including moving and stationary elements. CAVs can transmit vehicle information to roadside infrastructure and other vehicles. Similarly, connected roadside infrastructure and traffic management centers can transmit information about the current road conditions and traffic states. Based on these connected systems, a number of service packages (or applications) have been developed that provide specific safety and mobility-related services and benefits [1].

Despite the promise of this future connectivity and automation-based intelligent transportation systems (ITSs), the full potential of the massive data collection remains untapped. ITS encompasses technologies that aim to improve safety and mobility within the transportation system.

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ITS utilizes advanced communication technologies deployed within vehicles and transportation infrastructure [2]. Thus, ITS can combine data from vehicles, travelers and infrastructure to implement applications, such as vehicle-based traffic surveillance. However, while ITS offers advanced capabilities to monitor transportation systems and carry out certain applications [1], the true potential of connectivity and data collection is underutilized. It is envisioned that the concept of IoT, CPS, and DT technologies, which all are under the umbrella of ITS, could significantly improve safety and mobility within the transportation system.

IoT systems comprise of interconnected physical devices with embedded sensors and processing capabilities [7]. On the other hand, CPS systems connect physical objects with the cyber world and integrate feedback action based on the input from the physical objects so that the system can provide some output in the form of decision and actuation [3]. DT has been defined as “a virtual representation of what has been produced” [8]. Essentially, a DT creates a digital replica of a physical asset by utilizing the real-time state information of that physical asset object. DTs have also been defined as virtual models of physical objects that simulate behaviors, understand states through sensor data, and analyze dynamic changes, guiding the physical objects to optimize responses based on simulation insights [9]. DT systems can thus better utilize the data and perform predictive analysis of the transportation network using advanced predictive models, such as machine learning (ML) models, and simulations, such as agent-based simulation and microscopic traffic simulation environment. In this regard, DT technology can take advantage of this connected ecosystem and provide significant tangible benefits to the transportation system compared to IoT and CPS technologies.

It is important to note that a requirement of DT representation of an object is that there must be real-time synchronization of the virtual and physical objects through a frequent update of the state information. DT systems are often considered analogous to the IoT or CPS. While DT systems share components with IoT and CPS, there is a fundamental difference between these systems. DT systems are broader in scope and consideration and encompass IoT and CPS, but the same is not true the other way around. However, both IoT and CPS systems lack the concept of maintaining a virtual twin of the physical asset. Virtual twins are the digital versions of a corresponding physical asset of the DT system. As the virtual twin is intended to be a direct representation of the physical object, it can be used to perform predictive analytics that would otherwise be impossible to perform on the physical asset itself.

DT systems are finding applications across a wide range of industries [10]. In manufacturing, DT systems have been used for applications, such as maintenance, product lifecycle management, and production planning control [11], [12], [13], [14]. The aviation industry has found the use of DTs for design, assembly, manufacturing, operation, and maintenance of aircraft [15], [16], [17]. Electric power systems and smart grids also benefit from the adoption of DT technologies. Power system DTs offer power equipment malfunction detection,

load prediction, intelligent image inspection, power system analysis, and health state evaluation [18], [19], [20], [21], [22], [23]. Recently, in transportation, the proliferation of ITS technology has seen the widespread deployment of sensors and connectivity across the entire transportation ecosystem, which motivates researchers to demonstrate the efficacy and benefits of DT in the transportation domain.

B. Motivation

TDT systems extend the capability of ITS to implement proactive safety and mobility applications. Indeed, prototype systems targeting specific use cases are being developed to realize the benefits of a TDT system. Applications, such as cooperative ramp merging [24], cooperative driving [25], and driver behavior modeling [26], [27], have been developed to illustrate the efficacy of TDT systems in improving the safety and mobility of existing transportation systems. Liao et al. demonstrated a cooperative ramp merging application to facilitate the safe merging of vehicles powered by a vehicle-to-cloud DT. Their implemented solution could achieve a 67.41% reduction in speed variance in comparison to a baseline case that did not utilize a DT-based approach [24], [28]. Chen et al. [26] developed a DT-based safety application to avoid collisions between vehicles. Their solution first used DTs of drivers to predict their behavior in the driving context. Afterward, a risk analysis model was implemented that utilized the predicted driver behavior generated using shared DT models to determine the probability of collisions. Results demonstrated that the DT-based solution was able to achieve a significant reduction in collisions as compared to a scenario without the prediction algorithm and DT-based solution. Wang et al. [25] achieved a 20% reduction in travel time when a DT-based cooperative driving scheme was implemented at a nonsignalized intersection. Saroj et al. [29] developed the DT of a smart traffic corridor using real-time connected data from ITS data sources. Their solution enables studying how ITS adoption affects the corridor’s environmental performance and traffic flow. In another study [30], Saroj et al. demonstrated how such a DT enabled smart corridor could be used to optimize Reinforcement Learning based actuated traffic signal. The simulation results of their developed DT-based solution showed that it could achieve a 20.81% reduction in average travel times compared to the baseline scenario, which used an actuated signal. These recent studies effectively demonstrate the potential of TDT system implementation in improving safety and mobility within ITS.

In a TDT system, sensors are not only able to sense the physical processes and translate them into the digital domain but also data from these sensors can also be used to quantify human mobility behavior [31]. Historical crowd-sourced GPS location data can be processed to understand coarse-grained human mobility patterns in cities. Recently, it has been demonstrated that in a DT system, a more detailed understanding of human mobility patterns can be performed in the form of trajectory predictions [32]. Additionally, research has looked into the integration of humans into smart systems,

such as smart factories. As human characteristics differ significantly from the other elements in a transportation system, Sowe et al. [33] and Sahinel et al. [34] have discussed methods to model humans alongside other components and integrate them into the system considering human motivation, cognitive ability and predictability of task performance. Thus, such a system is also able to represent human mobility behaviors across groups, such as drivers and pedestrians.

Even in the backdrop of such interest in TDT systems, there is a lack of in-depth review of DT-based transportation safety and mobility applications and identifying challenges that arise from the requirements of such TDT systems. More importantly, there has been no consensus on developing a unified framework and architecture for realizing TDT systems. The majority of the implementations have developed architectures that are specific to their use cases. However, in order to ensure cross-compatibility across different TDT systems, it is essential to have an overarching reference architecture that can guide practitioners in developing TDT systems for any particular use case and at any particular level of transportation systems ranging from individual vehicles to city-wide DT systems of transportation networks. Also, it is necessary to identify future research directions stemming from the requirements of such a TDT framework.

C. Focus and Contributions of This Article

Although there are several published articles that have aimed to present a review in the transportation domain [3], [4], [5], [6], we intend to provide a holistic and comprehensive review by incorporating the gaps that are present in the other published review articles. Table I presents the survey articles that we found related to DTs in transportation systems. In one survey, Koscka-Olejnik et al. [5] reviewed DT systems in the context of internal transport systems. As the focus of that survey was related to internal transportation systems in manufacturing, the discussions were not sufficient for DT-based surface transportation systems, but there were some future research directions provided for DT research on transportation. Bao et al. [4] presented a literature review of DTs for ITS where they discussed DT systems in ITS, architecture for the DT system, their enabling technologies as well as applications of TDT. However, they had limited discussions related to TDT literature, challenges and future directions. Ali et al. [3] also reviewed DTs in ITS with a focus on providing EV services using DT approaches. On the topic of TDT systems, only DT in ITS and partial discussions on key technologies are presented. The most complete review of the literature on TDT was found in [6]. Additionally, they also elaborated on the applications of TDT systems, challenges, future research direction and review of the key technologies partially. However, it still lacked a comprehensive reference architecture for a TDT system as well as a discussion on DT in ITS. Therefore, in this review article, we identified these key areas and addressed each of them in light of the recent literature on TDT systems. Thereby, we address the gaps in previous survey papers related to TDT systems. The primary contributions of our article are the following: 1) conduct

an in-depth review of DT technology applications in the transportation domain, i.e., safety or mobility applications; 2) develop a reference architecture for a TDT system based on a hierarchical concept to guide in developing TDT systems at any scale; 3) identify research challenges for each component, i.e., the physical space, the communication gateway and the digital space of the TDT system, to realize safe and operationally efficient ITS; and 4) provide future research perspectives to develop and deploy networkwide TDT systems considering the scale and dynamic nature of future connected and automated transportation systems.

II. DIGITAL TWIN AND ITS RELATIONSHIP WITH TRANSPORTATION APPLICATIONS

The concept of DT in transportation systems is still in the nascent stage and consequently, a formal definition of a TDT system has yet to be formulated. Based upon the general definitions of DT systems, a TDT system may be defined as a virtual representation of transportation systems that maintain a digital replica of physical transportation system elements, such as vehicles, roadways, pedestrians. A TDT system must be a real-time representation of the corresponding physical transportation elements to mimic the properties, behavior, and interactions between various transportation elements.

The use cases for DT in transportation are not entirely new but rather have been envisioned within the ITS ecosystem through the ARC-IT framework [1]. ARC-IT provided a common framework to design and develop various traffic applications based on ITS capabilities. These applications, which include safety and mobility improvement applications, are defined as service packages within the framework.

An example of a safety application based upon the ARC-IT framework is the “Curve Speed Warning” service package. It transmits the recommended speed for a roadway curve to all approaching connected vehicles. Vehicles can utilize the information to detect if their approach speed is above the safe speed and warn drivers accordingly [1]. On the mobility front, the “Traffic Incident Management System” service package addresses mobility issues arising from traffic incidents. Roadside surveillance equipment is used to detect traffic incidents and once detected, the service package is able to coordinate with relevant agencies to prepare an apt response to the incident [1].

TDT systems will aim to extend the functionalities of these ITS applications based on the ARC-IT framework. In particular, TDT systems have the advantage of greater predictive capability and will be able to act in a proactive manner whereas the existing ARC-IT service packages are reactive in nature. For developing TDT systems, it is useful to have a scheme of categorization that can help to describe a TDT system. For this purpose, the following classification is presented here [35], [36], [37], [38].

- 1) *Product Lifecycle-Based Classification*: This classification of a DT system is based upon the concept of product lifecycle which describes the evolution of a product from conception to eventual withdrawal. The following two classifications may be performed under this.

TABLE I
COMPARISON OF EXISTING SURVEY LITERATURE TO OUR ARTICLE

Survey Source	TDT Architecture	Review of Existing TDT Literature	DT in ITS	Challenges	TDT Applications	Future Research Direction	Key Technologies
Ali et al. [3]	NP	NP	P	NP	NP	NP	PP
Bao et al. [4]	P	PP	P	PP	P	PP	P
Kosacka-Olejnik et al. [5]	NP	NP	NP	NP	NP	PP	NP
Schwarz and Wang [6]	NP	P	NP	P	P	P	PP
Our paper	P	P	P	P	P	P	P

Note: P=Present; NP=Not Present; PP=Partially Present

- a) *DTP*: The DT prototype is a digital version of a physical prototype that can be used to conduct tests that are not otherwise possible with physical prototyping. For example, a vehicle DTP may be used to model the dynamics of a specific type of vehicle. The vehicle DTP can then be used to run a large number of simulations to assess the crash safety of the vehicle. However, doing the same in the real world using a physical vehicle would not be feasible due to constraints of resources, such as time and funding.
- b) *DTI*: Once the physical component of the DT is produced and interfaced with it, a DT instance is created. DTI results when the DTP is interfaced with its physical twin version after the physical asset is manufactured. Each additional unit of the physical twin will result in an additional DTI.
- 2) *Value Proposition-Based Classification*: In this classification, a DT can be classified as follows depending upon the specific value it provides through its usage.
 - a) *SDT*: SDTs constitute the most basic versions of DTs and are limited to representing the state of the physical twin or process that is being replicated. These systems simply present information that the related personnel can use in making decisions and taking action. In a transportation context, an example of an SDT system is a real-time traffic visualization system for a roadway section. Such a system provides real-time visualization of the traffic situation through real-time video feeds and does not provide any interaction capability or predictive features. Examples of SDT-type implementations in transportation are [39], [40], [41].
 - b) *IDT*: An IDT, in addition to representing the real-time state of the physical object, is also able to control some aspects of those objects. By making control decisions based on real-time state information, an IDT is able to achieve better control than a human operator taking decisions based on information from an SDT. The real-time traffic visualization SDT system can be considered an IDT if the system integrates the ability to interact with surface vehicles. Such interactions can be in the form of advisory messages sent to variable message signs along the road. Such IDT examples are found in [24], [42], [43], and [44].

- c) *PDT*: This type is the most complex implementation of a DT. PDT systems are able to monitor the state of the physical object and use sophisticated algorithms and modeling to achieve proactive capabilities. An example of a PDT system would be a traffic rerouting application that optimizes the travel times for all vehicles. Such a system not only monitors the traffic but also performs extensive computations to predict the future state of the traffic and optimize accordingly. Furthermore, it is able to send out rerouting advisory information to the vehicles. PDT type implementations are seen in [25], [26], [27], [45], [46], [47], [48], and [49].

III. TRANSPORTATION DIGITAL TWIN ARCHITECTURES

This section presents TDT considerations, concepts and architecture for safety and mobility applications. According to [50], the most critical factors in developing DT systems are complexity, breadth, and depth. Complexity relates to the time and cost required to develop and implement the DT system. Breadth is determined by the level of generality or specificity required for the DT system. Depth, on the other hand, depends on the precision and accuracy required for the design and implementation of the system. Proper consideration of these factors is essential for the successful development of effective DT systems [50]. A hierarchical concept for the design and implementation of TDT systems, taking into account the considerations of complexity, breadth, and depth is presented in Fig. 1 which illustrates the concept, starting from the least complex and narrowest breadth at the individual asset twin level. At this level, DT systems are designed for discrete units, such as a single vehicle, pedestrian, or a traffic signal controller, with limited interaction with other elements of the transportation system. Although the breadth of the individual asset twin is narrow and specific to a particular vehicle model or pedestrian, high precision and accuracy are required for building any applications using the DT of the individual assets. The hierarchical concept progresses upward in complexity and breadth, as discussed in the following sections.

The level immediately above the individual asset twin is the network subcomponents twin, which includes the individual asset twins and the DT representation of subcomponents that make up the entire transportation network. Thus, in addition to the elements of the individual asset twins, this level incorporates the DT of roadway intersections, basic freeway segment, and various roadway geometric configurations, such

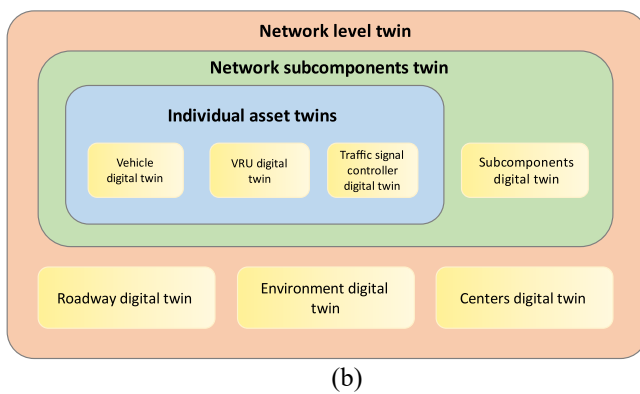
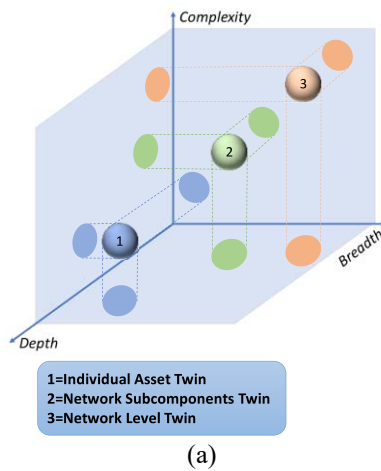


Fig. 1. (a) Complexity, depth, and breadth considerations within different levels of TDT system and (b) hierarchical concept for TDT systems.

as weaving segments, and on- and off-ramps. At this level, the interaction between individual elements becomes more relevant, and it is necessary to model their interaction in the digital space of the TDT system.

The most complex and broad level within the hierarchy is the network level DT. This represents a TDT system across multiple roadway network subcomponents forming a network. Thus, a network-level twin can be just several connected network subcomponents or as large as a city-wide system that combines the DT of each roadway network subcomponents. Hence, it will augment the intersection level twin through the incorporation of roadway, environment, and traffic center twins. Roadway twins will replicate the state of the roads and report parameters, such as traffic flow, density, and pavement health, and any traffic incidents that may occur. For a network-level twin, it is necessary to consider weather and other factors that may affect the flow of traffic. This prompts the addition of an environmental state at this DT level. Under traffic centers, many different services may be grouped together, including emergency and first responder services. The inclusion of centers within a DT representation of the network will enable the DT systems to coordinate signals across multiple intersections and optimize the route for such services to aid in incident or emergency response, thereby restoring mobility in the shortest duration.

Fig. 1(b) demonstrates the different levels within the hierarchy and their individual components. The organization of the proposed hierarchy enables the entire transportation system to be modeled at different levels. Since each of the higher levels within the hierarchy incorporates the lower level twins, they can inherit the functionality of those levels. Thus, a network level DT would be able to model city wide or regional level transportation systems. Additionally, as the lower level twins will be present, the modeling can be extended down to individual intersections or even at the level of individual vehicles, pedestrians or transit passenger. Consequently, applications, such as passenger demand modeling, can be performed at all levels up to the regional level [48].

Within the scope of the reviewed research on TDT systems, there has been numerous implementations of TDT for various applications. In the absence of a general architecture for TDT systems, these implementations have adopted proprietary architectures for their use cases. A three layer architecture was proposed in [46] which consisted of a field test layer, a network transport layer and the laboratory test layer. These represented the physical space, data communication link and digital space, respectively. This enabled them to implement the DT of an autonomous vehicle for testing within a virtual environment. Krämmer et al. [40] developed a DT of a section of a highway where they presented a combined hardware and software architecture for the system. They used camera and radar sensors and fused them in software using a dedicated data fusion unit. The fused data is used to create a visualization of the physical highway section [40]. Since this represents an SDT, their architecture lacked the components for data storage and also for doing any predictive analysis of the data. Furthermore, since the number of data sources was small, no data validation or missing data imputation was performed on the raw sensor data. A more comprehensive architecture was presented by Wang et al. [51] with four distinct layers: 1) the device layer; 2) the edge layer; 3) the application programming interface layer; and 4) the cloud layer. Compared to the previous architectures, their architecture was able to deal with real-time data ingestion as well as data storage and run a ML-based microservice for predictive analysis [51].

Considering the lack of a general architecture to support the development of TDT systems, we aim to propose an architecture in this section by augmenting the hitherto available TDT system implementations. Specifically, in order to facilitate the various functions in the digital space, some DT components and terminologies were introduced from other domains. These are discussed below.

- 1) *Digital Shadow*: In this stage, there is automated unidirectional data flow from the physical object to its digital counterpart. However, no data flow occurs from the digital version to the physical asset [36]. As suggested in the nomenclature, it is like a “shadow” of the real asset and only reflects the real-time state of the asset. Changing the digital state of the digital shadow will not induce any change in the state of the physical asset as there is no automatic data flow in the reverse direction [37].

- 2) *Digital Sibling*: Digital siblings aim to perform predictive analytics by constructing parallel simulated versions of the DT based on different cases and criteria. In this way, if one parameter in the physical world is altered, the effect on the state of the physical world and associated objects can be predicted. Hence, when data analytics and simulations are used to predict the future state of the physical asset, it can be considered a digital sibling of the asset [52]. It is a closely related “sibling” version of the physical asset and can be used to perform predictive analysis, optimization, etc. instead of the actual physical asset.

The fundamental parts of our proposed architecture consist of three distinct layers- the physical space, the DT, and the communication gateway layer. The synergistic coordination of the three layers enables the realization of the benefits of a TDT system. Fig. 2 illustrates our proposed reference architecture for the TDT. The following sections describe each of the architecture components in detail.

A. Physical Space

The components of the DT within the physical world have been highlighted by Wang et al. in their study [51]. In their mobility DT framework, they incorporated the human, traffic and vehicle elements into the physical space of the design. Rudskoy et al. [48] also presented a design for DT in ITS, where they integrated services for monitoring weather conditions, incident tracking, and traffic management along with the data for vehicle and road network. Consequently, a combination of these elements has been incorporated into our proposed architecture to enable holistic sensing and information flow across the transportation system.

The physical layer represents all the elements in the physical world and consists of the vehicles, ORUs, traffic management infrastructure, traffic control infrastructure, roadside infrastructure, roadway equipment as well as information from the nontransportation services like weather and emergency services. The transportation-related elements can be broadly grouped together into fixed and mobile nodes based on whether they can change their location over time. In the physical layer, the main functionalities that need to be provided are the sensing and actuation capabilities. In order to satisfy the requirements of a true DT, two criteria need to be ensured from the data standpoint. First, the data must represent the state of the physical object. Second, the state information must be a live representation, with the minimum possible communication delay between the actual state of the object and the sensed state [35]. Thus, within each component of the physical world, there needs to be a system of sensors that can satisfy the above-mentioned criteria and generate data for a DT system to consume.

- 1) *Mobile Nodes*: The mobile nodes of the physical transportation system comprise of vehicles, human drivers and ORUs. Within vehicle systems, internal sensors measure using various internal states as well as the motion states of the vehicle. The internal states of the vehicle contain crucial information related to vehicle speed, acceleration, braking and

steering angles. Other sensors can measure the proximity and speeds of nearby vehicles, ORUs and other surrounding objects in the roadway environment. Such sensors could include cameras, RADARs, LIDARs, ultrasonic sensors, and GPS units [53]. Whilst the sensor data from a vehicle could be used to implement a DT of itself within an in-vehicle computer [42], such a twin would not satisfy the requirements of a TDT. It is imperative that the real-time data from the vehicles be transmitted to the backend infrastructure where the transportation DT and all associated safety, and mobility can be implemented. Data from individual vehicles is especially important for developing and implementing TDT-based traffic safety applications. Apart from the road vehicles, ORUs, such as pedestrians, wheelchair users, and people using nonmotorized vehicles also represent mobile nodes as they can change their location over time. Typically, ORUs are much slower moving as compared to vehicles, and the relevant state information for them is location information. Approximate location information of ORUs, accurate to within several meters, can be obtained from the location information transmitted by personal handheld wireless devices like smartphones [54]. In order to obtain greater accuracy of ORU location within a specific segment of the road, roadside infrastructure, such as cameras, RADARs, and LIDARS, may be utilized [55].

- 2) *Fixed Nodes*: In contrast to the mobile nodes, the fixed nodes of the transportation system are the stationary physical assets comprising of traffic control infrastructure, roadside infrastructure, roadway equipment, and traffic management center. Traffic control infrastructure, such as traffic signal controllers, are fundamental to ensuring the mobility of transportation systems. The signal phase and timing information of the deployed signal controllers and the current updated phase information of the signal controllers will need to be monitored and replicated in the DT versions to construct the virtual representation of the physical controllers. Roadside infrastructure and roadway equipment are the traffic infrastructures placed on the side of the road and embedded within the pavement, respectively. Thus, roadside equipment could include sensors, such as cameras, LIDARs, RADARs, wireless communication radios, and low-powered (i.e., edge) computing devices. These can monitor the traffic states on a roadway segment or intersection, thereby providing real-time data such as traffic volumes and vehicle trajectories on the road [56], [57], [58], [59]. Roadway equipment, such as inductive loop detectors, can also sense the roadway occupancy information. Other roadway equipment includes future wireless charging systems embedded within the pavement for wirelessly charging EVs while in motion. DT can be used to optimize such a charging system and hence the data from these systems can be used to construct a DT version for monitoring and optimization.

- 3) *Nontransportation Services*: The nontransportation services include all such services that are not directly part of the transportation system but can indirectly influence the mobility and safety of the system such as weather services and emergency services. In contrast to the other components, these services will only send the relevant information to the DT and will receive no feedback or output from the backend. Adverse weather conditions can impact mobility by causing

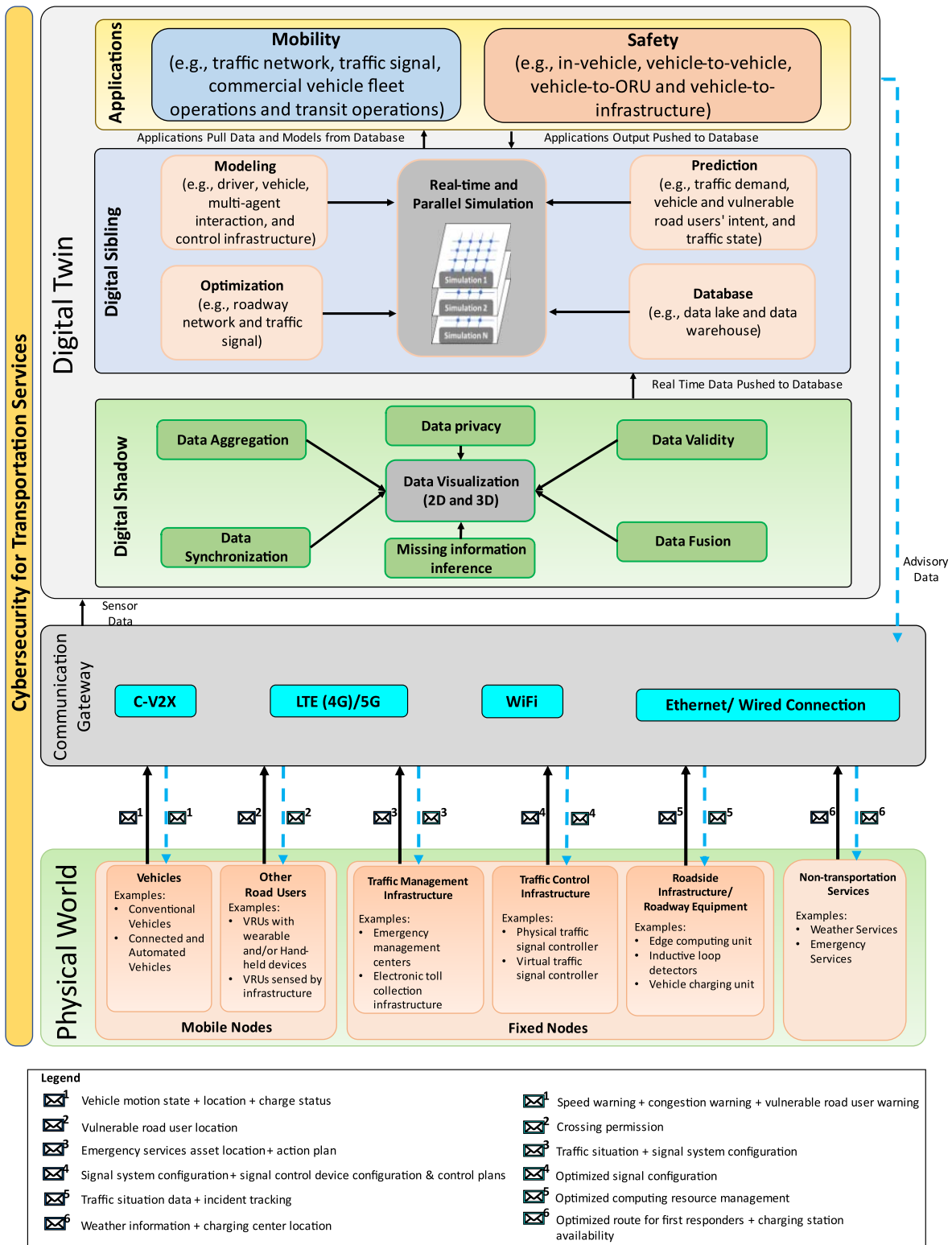


Fig. 2. Reference architecture for TDT.

damage to transportation infrastructure or making it hazardous to operate vehicles. Hence, it must be accounted for and replicated in the DT version of the transportation system to enable accurate analytics to be performed. In addition, emergency services like police, ambulance and first responders have very specific mobility demands due to the nature of their

operation. Information from the related centers also needs to be transferred to the DT.

4) *Actuation*: Apart from the sensing function, actuation functionality is also required to be provided in the physical space. Actuation in this context can take the form of receiving and acting upon advisory messages from the digital space,

such as suggested routes to avoid congestion, or it can be in the form of actual change in the state of an object, for example, changing the traffic light phase. The actuation capabilities enable the TDT to implement the outputs of its safety and mobility applications.

B. Communication Gateway

The communication gateway is an integral part of any IoT, CPS or DT system. Although in prior studies [51], the communication layer has been mentioned as a part of the DT architecture, there has been no discussion on the requirements for communication and the available communication options. As characterized in [60], the communication between the physical asset counterpart and the digital version is referred to as intratwin communication. For intratwin communication, C-V2X has been mentioned in [51], and 4G/LTE has been used by the Liao et al. [24] for their cooperative ramp-merging application. Additionally, 5G has been tested in [49], for implementing a vehicle DT within a broader system that involved a DT of a highway section.

As the interface between the physical space and the digital space, the communication gateways facilitate the transmission of data between them. Data from the fixed and mobile nodes as well as nontransportation services are pushed to the digital space and actuation signals from the digital space are routed to the relevant physical asset. There are a multitude of wireless communication options that can be adopted for use within this layer. However, the DT system requires a number of requirements to be fulfilled by the communication technology that is used. First, it must have a high-bandwidth and high-data rate capability. Second, there must be minimum latency or delay in the transmission of data and information. Finally, as the system will consist of a large number of data sources, the communication technology must be able to be easily scalable as required.

Communications in DT systems can be classified into three categories: 1) P2V; 2) P2P; and 3) V2V [61]. P2V communications refer to when a physical object uses wired or wireless communication technology to communicate with a virtual twin, sharing the physical object's data in real time and receiving input from the virtual twin. Here, wide-area network wireless communication technology, such as LoRa and 5G/6G cellular communications, is mostly used. The 5G-based communication has a range of up to a 100km and data rates of about 599Mb/s which is the highest among all wireless communication options. Although LoRa has a much more limited range and throughput compared to 5G, it does not require a license to operate which can help to ease regulatory requirements in real-world deployments. Their great ranges make them ideal for deployment in network-level twins. In this case, the physical object is a wireless terminal that is linked to the wireless access network via a wireless communication base station and then to the virtual twin on the Internet. P2P communications enable information sharing between physical objects. Furthermore, a variety of communication protocols, such as the WPAN, ZigBee, and LPWANs technologies, such as LoRa and NB-IoT enable network connectivity [61].

TABLE II
PROPERTIES OF VARIOUS WIRELESS COMMUNICATION PROTOCOLS AND THEIR APPLICATION TO DT USE CASES

Wireless communication protocol	Frequency spectrum	Range	Data rate	Potential DT use case
LPWAN	2.4 GHz	25-50 m	50 kbps	Individual level twin
ZigBee	Unlicensed subGHz	10-100 m	250 kbps	Network subcomponents twin
NB-IoT	Cellular band	1-15 km	250 kbps	Network level twin
5G	Cellular band	Up to 100 km	599 Mbps	Network level twin
LoRa	Unlicensed subGHz	2-15 km	50 kbps	Network level twin

Although NB-IoT uses the licensed cellular bandwidth, ZigBee operates on an unlicensed spectrum while LPWAN uses the 2.4-GHz spectrum used ubiquitously in Wi-Fi devices. For P2P communications under 100m, LPWAN and ZigBee provide a sufficient range, but for longer distances, NB-IoT is more viable. All three options are limited to low-data rate transmissions between 50–250 kb/s. Hence, considering their range and data rate, they are mostly suitable for use in network subcomponents twins. Table II summarizes the properties of the available wireless communication options and potential DT use cases [62], [63]. V2V communications occur in the virtual space. It imitates the communication behavior seen in the actual world. In a transportation DT context, V2V communications represent the data transmission between the DT model entities of the vehicles.

Unlike real vehicle communications, which use wireless spectrum and radio power, this virtual mode relies heavily on the processing capacity of DT servers to replicate data transfer behavior [61]. Apart from the wireless network technology, the choice of messaging protocols is also important in DT. Due to the low-latency requirement, lightweight messaging protocols are preferred instead of traditional http protocols for the transmission of data to the cloud. Here, two candidates are the most preferred: 1) the MQTT protocol and 2) the CoAP [61].

C. Digital Twin

This section describes all the digital components of the DT block of our architecture. Real-time data from the physical space serves as the input to the digital processes and the output is actuation commands or advisory messages. In our architecture, the three main constituents for the DT are: 1) the digital shadow; 2) the digital sibling; and 3) applications. Although in other domains [11], [64], the digital shadow and digital sibling have been distinguished from the actual DT by their definitions, in our architecture we have incorporated them within the DT without violating their defining criteria. This design choice was made to make the architecture as general as possible. This enables flexibility in the design and implementation of an actual TDT system. For example, if the design requirements do not necessitate any predictive

functionalities as is the case for a SDT or IDT, it can be chosen to exclude the digital sibling component and only include the digital shadow and applications components. Each of these components has a distinct function which is detailed below.

1) *Digital Shadow*: The functionalities within the digital shadow component have been implemented across various traffic safety and mobility applications using DTs. Saroj et al. [65] implemented missing data imputations for the case of traffic volume data. Other functionalities, such as data aggregation, synchronization, and validation, have been described in [66], [67], and [68]. State information for the physical elements are transmitted to the backend or cloud systems that host all the digital elements of the TDT. A number of operations are then performed on the data in order to create the DT representations. Data from disparate sources may need to be aggregated via data aggregation operations in order to have a single source from which required data may be pulled as required. Often this aggregation operation pools data into repositories called data lakes and data warehouses. In particular, data lakes allow the storage of raw structured, semi-structured or unstructured data and have the advantage of being easily scalable to meet the demands of real-time operations [69]. Hence, for real-time digital replicas, data lakes provide an ideal solution for aggregating the data. Another important operation on the data that needs to be performed is the synchronization of sensor data across multiple sensors. Within the physical space of the TDT, sensors will also be spatially distributed throughout the entire transportation network. This can induce variable data transmission delays due to the differences in transmission path length. Also, it is unlikely that all the sensors and data sources will have the same sampling rate or data frequency. Thus, in order to make this data useful for real-time applications, the aggregated data needs to be synchronized in a common timeframe prior to using it for the state representation of an object. This will ensure that the virtual state representations are synchronized with the physical states of the object that is being replicated [70]. Data from a single sensor modality may not be adequate to determine the state of the physical object due to the inevitability of factors, such as noise, occlusion, sensor failure. In such instances, data fusion techniques can help to improve the precision and accuracy of a particular sensed value by combining sensor data across two or more sensor modalities. The data being sourced from the real-world sensors needs to be validated before being utilized by models within the DT and this can be ensured by the data validation block within the digital shadow. This data may also suffer from missing information. In order to work around this, imputation needs to be performed to determine the missing data. Since data from individual level twin would consist of vehicles and ORUs, such as pedestrians and transit passengers, the data from them might contain privacy sensitive information. Hence, the digital shadow will also need to perform operations related to data privacy to ensure all personally identifiable information is removed.

2) *Digital Sibling*: The second component of the TDT architecture is the digital sibling. The digital sibling component performs all the computation tasks that are required to implement the safety and mobility applications of the

TDT. Thus, the major functionalities of this component are modeling, simulation, prediction, optimization and database operations. Modeling using DT has been described in [71], to model the interaction of a vehicle tire and road system. Other studies [30], [72] have implemented optimization using their DT system for vehicle frame mass optimization and traffic signal optimization. A DT system to predict driver behavior has been implemented in [27] while on the macro-scale, DT systems have been used to predict traffic in [25] and [73]. A parallel simulation-based approach was used to improve software defined vehicle networks using DT capabilities in [74]. Database requirements for TDT systems and transportation data requirements have been discussed in [27] and [51]. Modeling involves constructing relevant models of the physical object, which can enable interaction with the objects in simulation environments. Thus, geometric modeling, physics-based modeling and data-driven modeling are utilized here to enable interaction with the digital replicas of the assets. Whilst the availability of real-time data in TDT systems would make it easier to develop and train data-driven models, physics-based modeling can also complement them. Aside from the modeling of individual assets, it will also be necessary to model the specific transportation services that will need to be implemented. The developed models can be used to implement specific simulation scenarios of traffic environments. As a precursor to DT system, MAS enabled the study of independent intelligent agents coexisting in a common environment. The agents in the MAS can interact with the environment and with each other which enables the understanding of the dynamics of the system [75], [76], [77]. In many ITS applications, MAS is used to evaluate the effect of agent interaction with the ITS system [78], [79], [80], [81], [82].

DT systems fundamentally differ from such MAS implementations in that the models and simulations within DT systems consume real-time or near real-time data from physical sensors [83]. The data from these sensors are used to update the simulations and model the evolution of the system thereby. In contrast, MAS only utilizes the initial state of the system and subsequently, the simulation evolves based on the agent-agent and agent-environment interactions. A DT system can incorporate a MAS in the digital representation of the physical asset, but it must be augmented by the injection of real-time data. In that respect, a DT overcomes a limitation of MAS which is the divergence of the simulation from the current state of the physical system. The simulation can use digitally modeled elements to observe the interaction between these elements in the digital space without the interactions ever taking place in the physical world. For example, parallel simulations can be performed to evaluate the outcomes of routing decisions and identify the best choice thereby. Each simulation will track the outcome of a particular routing decision to see the final impact on improving the mobility of the system. For safety applications, the digital sibling component can simulate interactions between ORUs and vehicles to identify potential conflicts and accident-prone zones. This enables the TDT to proactively identify safety and mobility issues and suggest measures for correction. Thus, prediction tasks are another functionality of the digital sibling component. Furthermore,

the digital sibling can also enable the optimization of the models and simulations. Finally, the database function within the sibling enables it to store structured and unstructured data. In this regard, data warehouses are better suited for use in the sibling [69]. Data warehouses can store large amounts of data which, unlike data lakes, must be stored in a structured form. This makes it appropriate for use as a historical data store, as it is often required to perform complex data queries on such data. Historical data related to transportation and nontransportation systems can be pulled from the database. Through the comparison of the prediction results with the historical data, it would be possible to optimize the models used in prediction as well as any tunable simulation parameters. This can refine the models and increase their fidelity over time enabling higher accuracy of prediction. The outputs of the training and simulation can be pushed to the database to save them for future purposes as required.

3) *Intertwin Communication*: In ITS systems, as referenced within the hierarchical structure of the TDT system (see Fig. 1), higher order systems (i.e., network level) can build upon the lower order systems (i.e., network subcomponents level and individual asset level) in order to construct the whole TDT system. In this regard, an important consideration is the communication between disparate DT or intertwin communication [84]. As opposed to intratwin communication [84], which transfers data between the physical version and the virtual DT representation, the intertwin communication enables the exchange of data and models between different twins within the virtual world. Thus, a physical asset could be spatially separated from another but can exchange data and interact with each other through intertwin communication in the DT layer. The intertwin communication can emulate communication between the physical assets, facilitate communication between collaborating groups of DTs, or enable communication between physical assets that cannot otherwise be implemented in the physical world due to various constraints [85]. Intertwin communication is an important aspect for TDT implementation within ITS as it enables entities like vehicles to communicate beyond their range of communication in the real world, which would otherwise take place through a wired or wireless communication media, such as Cellular Vehicle-to-Everything (C-V2X). Through intertwin communication, only specific information may be exchanged while preserving privacy [60].

4) *Applications*: The applications block deals with the applications (safety or mobility) that are to be implemented using the TDT system. It combines the functionalities of the digital sibling to accomplish the specific safety or mobility application that is desired. The outputs of the application may be used to drive an actuator in the real world or change the state of a physical asset. Additionally, the output may be pushed back to the digital sibling for storing as historical data.

IV. TDT APPLICATIONS REVIEW

The goal of implementing a TDT system would be to improve the transportation system related to safety, and mobility impact of the system. Indeed, the body of research

available in this realm has been directed toward exploring the applications of a TDT system and developing use cases for the technology. Therefore, in this section, the prevalent literature on TDT system applications and use cases is reviewed and presented. Table III presents the summary of the review. For each of the reviewed publications, we summarized the three individual components, i.e., the physical world, communication and DT component of the DT systems according to the level of the implementations. We have used three different terms to denote the levels: 1) conceptual; 2) simulation and 3) real world. We used the real world to describe components that have utilized real-world hardware and software components whereas simulation has been used to refer to components that have only been simulated using simulation software packages. And conceptual implementation is used to describe cases where the authors have only provided a conceptual description or formulation of their system component without undertaking a simulation or real-world implementation.

A. Safety

Traffic safety applications aim to improve the safety aspects of transportation systems. A TDT system can be used to implement specific applications to improve safety across interactions between vehicles and vehicles, and vehicle and ORUs. To address vehicle safety issues, Alam and El Saddik [42] presented a DT-based architecture for cloud-based CPS for a prototype ADAS. The prototype utilized sensor fusion on mobile sensor data and data from on board diagnostic II (OBD2) port of a vehicle to detect driving events in real-time. However, the presented prototype was purely conceptual, and no implementation was performed. Furthermore, the prototype did not consider the communication aspects of building the DT system, which is an essential component. Schellmann et al. [49] demonstrated a prototype for a 5G-based connected automated vehicle in an infrastructure-enabled DT of the road system. Their developed solution was able to provide real-time sensory information to a roadway DT system and also receive information from the system to improve automated driving functions. In [46], they implemented the DT of an autonomous vehicle and tested a real automated driving vehicle within a virtual test scene. Notably, their work demonstrated the use of V2X technology in a TDT application. A DT of a vehicle's braking system was developed in [43] using virtual and physical simulation technology. The advantage of their simulation method was that it had no dependence on any manufacturer-specific hardware. Almeaibed et al. [45] developed a conceptual framework for vehicular DT. They also developed a case study to validate the use of the vehicular DT framework to detect a cyber attack on a vehicle's radar sensor but did not carry out a physical demonstration of the attack and attack detection.

Further research has focused on human driver modeling through DT to improve safety. Kumar et al. [27] developed a DT application for driver intention prediction. Their approach involved a virtual model of a vehicle called a virtual vehicle, which contained the state of both the vehicle and the

driver. They evaluated the success of their approach by achieving successful vehicle coordination amongst a large number of vehicles without central control. In a similar theme, Chen et al. [26] proposed a framework for digital models of drivers and sharing them amongst connected vehicles to increase driving safety. A case study for highway driving was presented and the authors showed that their model could help to learn drivers' behaviors and enable crash prediction. Wang et al. [44] leveraged the Unity game engine to develop the DT of a CAV. They presented a case study of a personalized adaptive cruise control system that used a human-in-the-loop simulation to suite an adaptive cruise control system to a driver's individual preference for safety and comfort. Liu et al. [86] performed sensor fusion of cloud DT information with camera data to enable behavior prediction of a target vehicle. A case study was shown with a human-in-the-loop simulation where they found that their solution increased time-to-collision and reduced speed variance. However, a real-world evaluation of the safety benefits of the system was not performed.

As discussed earlier, the most complex level of TDT is a network-level twin. Several publications have tackled this challenge. A conceptual framework for the DT of a road surface was presented in [71]. Although the authors reviewed current enabling technologies for the road surface DT system, an actual system implementation was not presented. Similarly, Rudskoy et al. [48] discussed a DT architecture for implementing various services within an ITS system without showing a real system. In terms of real-world implementation of a roadway TDT system, Krämmer et al., [87] developed an architecture for a DT representation of a section of a highway. The system was implemented as an intelligent infrastructure system that could communicate with CAVs to improve their range of perception. The authors reported remarkable accuracy of the system in terms of classification and spatial position estimation which is promising for future network-level TDT implementations.

Other DT-based applications have been researched that improve the safety of transportation indirectly. Adibfar and Costin [88] demonstrated a bridge safety application that leveraged data from ITS sources and bridge information model (BrIM). However, the system was developed for a mock-up bridge system and only served as a proof of concept. Korostelkin et al. [72] used a DT of a vehicle to optimize the frame mass of a vehicle to meet the requirements of crash safety specifications. Their findings would be useful in improving safety at the individual vehicle asset level. In another study, Sahal et al. [89] proposed a framework to enable collaboration between DTs using a blockchain approach. Furthermore, they discussed how the collaborative framework may be applied to smart transportation use cases. Piromalis and Kantaros [90] discussed the application of DT technology in the automotive industry. They discuss the usage of vehicular DTs to improve safety and provide ADAS capabilities.

B. Mobility

Whilst safety applications improve the safety of the road system, mobility applications aim to improve and facilitate

the overall mobility of the system. TDT systems offer a great advantage in this regard as the optimization capability will allow the system to optimize traffic flow and reduce congestion. Similarly, other works have investigated mobility applications using DT systems. Barosan et al. [39] developed a DT simulation for autonomous truck operation at a distribution center in which they tested several kinematics models and path controllers. The developed simulation environment could be extended for use in studying truck platooning applications in a virtual environment. el Marai et al. [41] developed a method to create the DT of road infrastructure for smart city applications. The performance results showed that the system could detect and recognize objects, such as vehicles and people. Although no specific mobility applications were implemented, the system's ability to detect vehicles could be used in mobility-based DT applications. Dasgupta et al. [91] demonstrated a TDT-based solution to improve ATSCs performance. In their simulation of the DT system, the authors found that DT-based ATSC outperformed the connected vehicle-based baseline ATSC in terms of average cumulative waiting time, distribution of drivers' waiting time, and level of services for each approach for different traffic demands. This demonstrates how DT-based approaches can further improve existing mobility solutions.

In another paper, Wang et al. [25] demonstrated a cooperative driving scheme that enabled vehicles to pass through the intersection safely using a DT of an intersection. Advisory messages from the intersection DT instruct the drivers to adjust their speeds such that the vehicles do not have to stop before proceeding. Since no stoppage is necessary for the scenario, there is a significant reduction in travel time and energy consumption, thereby improving overall mobility. Saroj et al. [29] demonstrated a DT system for a smart traffic corridor where they fed real-world data from the corridor into a VISSIM simulation to study traffic flow. Their system could provide travel time, energy consumption and emission estimates but suffered drawbacks, such as missing data and computation bottlenecks, under heavy traffic demands. As an extension of this, they investigated the effect of data imputations on the performance of the DT [65]. They found that data imputation was more applicable to short-duration data gaps. However, their approach only considered errors due to data loss and did not address sensor errors. In [92], they developed a traffic flow prediction model based on DT architecture and the LSTM model. While their model showed promising performance in predicting short-term forecasts, it could not predict traffic for medium- or long-term scenarios. In [73], they also attempted network traffic prediction by using a reinforcement learning-based approach and found that their method is suitable for longer term traffic flow prediction. Xu et al. [93] developed a DT for mobility management within smart cities using a cloud-based microservices architecture. A radar-camera fusion-based scheme was developed to create a DT of a roadway section in [94]. DT applications have also been explored to enhance EV mobility. Zhang et al. [95] proposed a DT of a real-world EV that simulates the behavior and route choice of EVs based on time-varying driving operations, travel intent, and charging plan. Their DT system can

TABLE III
TRANSPORTATION DT APPLICATIONS

Application type	Study	Value based classification	Twinned entities	Implementation	Safety and Mobility Contributions
Safety	Kumar et al. (2018) [27]	PDT	Roadway and vehicles	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Utilized a DT approach for Driver intention prediction and congestion avoidance
	Chen et al. (2018)[26]	PDT	Driver	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Driver behavior prediction to support ADAS systems
	Liao et al. (2022)[24]	IDT	Driver and vehicle	- Physical Space: Real world - Communication: Real world - Digital Twin: Simulation	DT based Advanced Driver Assistance System (ADAS) to enable cooperative ramp merging
	Sahal et al. (2021)[89]	PDT	N/A	- Physical Space: Conceptual - Communication: Conceptual - Digital Twin: Conceptual	Proposed a concrete ledger-based collaborative DTs framework
	Ge et al. (2019) [46]	PDT	Vehicle	- Physical Space: Real world - Communication: Real world - Digital Twin: Simulation	Implemented a DT of an autonomous vehicle
	Piromalis and Kantaros (2022) [90]	PDT	Vehicle	- Physical Space: Conceptual - Communication: Conceptual - Digital Twin: Conceptual	Presented a discussion of DT technology application to the automotive industry
	Rudskoy et al. (2021)[48]	PDT	N/A	- Physical Space: Conceptual - Communication: Conceptual - Digital Twin: Conceptual	Presented a DT architecture for implementing various services within ITS
	Buechel et al. (2019)[49]	PDT	Vehicle	- Physical Space: Real world - Communication: Real world - Digital Twin: Simulation	Prototyped a 5G-connected and automated vehicle for use in an infrastructure-enabled DT system of the road traffic
	Almeaibed et al. (2021)[45]	PDT	Vehicle	- Physical Space: Conceptual - Communication: Conceptual - Digital Twin: Conceptual	Developed a framework for vehicular DT
	Krämmmer et al. (2022) [87]	SDT	Roadway and vehicles	- Physical Space: Real world - Communication: Real world - Digital Twin: Simulation	Implemented the DT of a section of highway
	Korostelkin et al. (2019)[72]	PDT	Vehicle	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Used a DT of a vehicle to optimize the frame mass of a vehicle for satisfying requirements of collision safety
	Dygalo et al. (2020)[43]	IDT	Vehicle	- Physical Space: Real world - Communication: Simulation - Digital Twin: Simulation	Implemented DT of a braking system of a vehicle
	Liu et al. (2020)[86]	PDT	Vehicle	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Proposed a sensor fusion method to fuse in-vehicle camera data with cloud DT information
	Kaliske et al. (2021)[71]	PDT	Road surface	- Physical Space: Conceptual - Communication: Conceptual - Digital Twin: Conceptual	Presented an overview of the requirements for building DT of a road system
	Wang et al. (2021)[44]	IDT	Vehicles, road networks	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Developed a DT simulation to present a case study of a personalized adaptive cruise control system
	Alam and El Saddik (2017)[42]	IDT	Vehicle	- Physical Space: Conceptual - Communication: Conceptual - Digital Twin: Conceptual	Presented a DT architecture for cloud-based cyber-physical system and a prototype for an advanced driver assistance system
	Adibfar and Costin (2022)[88]	IDT	Bridge, Vehicles	- Physical Space: Real world - Communication: Simulation - Digital Twin: Real world	Developed a DT based bridge safety application by integrating ITS and Bridge Information Model (BrIM) data

be used to study various algorithms related to EV navigation and charging within a simulation environment which can be used to inform charging recommendations and deployment plan of EV charging piles. Saroj et al. [30] implemented a DT system and application to optimize traffic signal timings in a smart connected corridor testbed. They reported a 20.81% reduction in travel times compared to actuated traffic controls that were running in the test bed. An approach to integrate real time traffic data into simulations was demonstrated in [97]. Additionally, a real-time analysis of their system was

presented on a section of a Swiss motorway to validate their approach.

To improve computation capability within vehicular networks, Yuan et al. [96] developed a framework to enable vehicular task offloading within the IoVs. Specifically, they used a DT-driven IoV along with Sixth Generation (6G) wireless network technologies, such as intelligent reflective surfaces, to build their task offloading system. The resulting framework was able to reduce the latency and energy consumption for vehicular task offloading in IoV. Xu et al. [98]

TABLE III
(CONTINUED) TRANSPORTATION DT APPLICATIONS

Mobility	Barosan et al. (2020)[39]	SDT	Vehicle	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Developed and implemented a DT for autonomous truck driving at distribution centers
	Dasgupta et al. (2021)[91]	IDT	Traffic signal controller	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Developed a DT based Adaptive traffic signal controller to minimize traffic delay in a congested situation
	Wang et al. (2022)[25]	PDT	Vehicle, Intersection	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Developed a DT framework for enabling a cooperative driving system at a non-signalized intersection
	Marai et al. (2020)[41]	SDT	Road infrastructure	- Physical Space: Real world - Communication: Real world - Digital Twin: Real world	Proposed method of creating DT of road infrastructure for smart cities
	Zhang et al. (2019)[95]	IDT	Vehicular edge network	- Physical Space: Real world - Communication: Simulated - Digital Twin: Simulated	Proposed DT-based modeling to conduct replication of the charging and discharging process of large-scale mobile EVs in different kinds of dynamic scenarios
	Saroj et al. (2021) [29]	SDT	Vehicles, traffic flow	- Physical Space: Real world - Communication: Real world - Digital Twin: Real world	Developed a DT traffic simulation model for a connected corridor
	Tu et al. (2022) [92]	PDT	Vehicles	- Physical Space: Real world - Communication: Conceptual - Digital Twin: Conceptual	Combined DT technology with LSTM network to enable short-term traffic flow prediction
	Saroj et al. (2022) [65]	IDT	N/A	- Physical Space: Real world - Communication: Conceptual - Digital Twin: Conceptual	Investigated the effect of data imputation on performance improvement of a transportation DT system
	Saroj et al. (2022) [30]	PDT	Vehicles, traffic signal controller	- Physical Space: Real world - Communication: Real world - Digital Twin: Real world	Optimized traffic signal timings in a smart connected corridor
	Li and Zhang (2023) [94]	SDT	Vehicles	- Physical Space: Real world - Communication: Real world - Digital Twin: Real world	Developed a DT of traffic flow using radar-camera fusion
	Xu et al. (2023) [93]	SDT	Vehicles, road infrastructure	- Physical Space: Real world - Communication: Real world - Digital Twin: Real world	Developed and implemented a DT system for mobility management in smart cities
	Kusic et al. (2023) [97]	PDT	Roadway	- Physical Space: Real world - Communication: Real world - Digital Twin: Real world	Developed a DT of a roadway section and integrated simulation with real-time data
	Nie et al. (2023) [73]	PDT	Transportation big data	- Physical Space: Real world - Communication: Conceptual - Digital Twin: Simulated	Proposed a reinforcement learning based approach for network traffic prediction
	Ji et al. (2023) [112]	SDT	Vehicle trajectories	- Physical Space: Real world - Communication: Simulation - Digital Twin: Simulation	Developed a framework for tracking vehicle trajectories using IoV communication to enable intersection DT
	Yuan et al. (2022)[96]	PDT	Vehicles	- Physical Space: Simulation - Communication: Simulation - Digital Twin: Simulation	Improved vehicular computation capability using intelligent reflective surface technology and edge computing in a DT vehicle network
	Xu et al. (2022)[113]	PDT	Vehicles	- Physical Space: Real world - Communication: Simulated - Digital Twin: Real world	Implemented a computation offloading scheme using decision theory in an ITS with a DT representation
	Hu et al. (2022)[99]	PDT	Traffic Flow	- Physical Space: Real World - Communication: Simulated - Digital Twin: Simulation	Developed a method for traffic flow and velocity prediction using data from a 5G enabled DT of a transportation system
	Feng et al. (2023)[100]	PDT	Traffic Flow	- Physical Space: Real World - Communication: Simulation - Digital Twin: Real World	Developed an algorithm to reduce adverse effect of uncertain conditions on IoV in transportation system

also proposed a method to implement vehicular task offloading in ITS, which utilized DT technology in conjunction with decision theory to make optimal offloading strategies. Hu et al. [99] developed a method for the prediction of traffic flow and velocity information using the real-world data from a DT system which is transmitted using 5G networks. Their method aimed to solve the important issue of missing data

from IoV sensors that hinder the applicability of DT data for prediction or optimization tasks, thereby contributing to the improvement of mobility of DT-enabled ITS. Feng et al. [100] studied how resilience of DT can affect the applicability of transportation systems. They developed an algorithm to improve the ability of the transportation network to mitigate the impact of external uncertainty.

V. RESEARCH CHALLENGES TOWARD TDT

As research interest in the area of TDT grows, it is evident that future transportation systems could see widespread deployment of DT technology. However, moving in that direction will require existing ITS to evolve. The presented TDT architecture (see Fig. 2) aims to generalize the design and implementation of such systems. Nevertheless, due to the current state of the art, there remain important challenges pertaining to each component of the architecture. It is necessary to note that most of the key challenges are not unique to TDT systems but pervasive across DTs in other domains as well. Fig. 3 depicts the specific research challenges pertaining to each part of the architecture presented in Fig. 2. The remainder of the section discusses these challenges in detail.

A. Physical World

In the physical space of the DT, the main challenges that are persistent are due to the gaps and limitations in current sensor technology. The precision and fidelity of the sensors dictate the performance of the digital models to a large extent. Sensors can suffer from varying levels of noise from different sources due to changes in the environment within transportation systems. This hinders accurate and precise measurement of the environmental variables [101], [102]. Cameras suffer from variations in lighting conditions throughout each day and also across seasons. LIDAR and RADAR-based sensors may be blinded by occlusions [103], [104]. The sensors are also prone to loss of calibration over time which adds to maintenance requirements [40]. Another challenge is the inherently mobile nature of transportation elements, such as vehicles and pedestrians [51]. Moving targets are difficult to track and although current algorithms are improving in terms of accuracy, there is still room for considerable improvement. Another issue with the sensor deployments in DT systems is the cost of implementation. As a large number of sensors are required to be deployed across the transportation system, the cost of the sensors, their installation, and maintenance can quickly become prohibitively expensive [105]. Many sensors produce enormous amounts of raw data which consume a significant amount of bandwidth to transmit across communication channels [106]. For such sensors, it is necessary to reduce the bandwidth consumption during transmission by filtering the data. The method of Compressed Sensing can be implemented to reduce the bandwidth requirement. Compressed Sensing enables the reconstruction of a signal with less samples compared to traditional sampling by leveraging the sparsity of a signal [107]. Therefore, sensor deployments need to have sufficient computing capability in the vicinity of the sensors to implement this filtering. Future research into new types of sensors can help to overcome the drawbacks of current sensors. Multispectral cameras can capture multiple wavelengths of light within the same camera unit. This enables it to capture more information that may not be obtained through regular standalone RGB cameras or infrared cameras [108], [109], [110]. As a result, multispectral cameras can overcome the limitations of individual camera types due to noise, range, or resolution limitations. Apart

from the deficiencies in sensing, challenges also exist in the incorporation of data from nontransportation services into the TDT system. These data need to be interoperable with the data from the fixed and mobile nodes. Furthermore, data, such as weather and traffic incidents, have significant uncertainties and errors.

B. Digital Twin

This section presents research challenges related to different components of DT.

1) *Digital Shadow*: The ability to visualize real-world data is one of the most basic propositions of any DT system. The visualization of data is an ongoing effort that has left much to be explored. While traditional data visualization relies on data dashboards, newer methods of human-computer interaction promise to revolutionize this. Some of these techniques are: augmented and virtual reality, natural language processing, and gesture control [52]. These technologies can enable more intuitive and natural interaction between the TDT system and the human user. However, the implementation of these technologies for DT system data visualization is still in its infancy and needs more research attention. Data fusion becomes a challenging task when performed at the scale of a transportation system due to the immense number of fixed and moving targets involved [40]. Furthermore, transportation systems are highly dynamic. Therefore, newer fusion methods that incorporate multimodal sensors need to be investigated. Specifically, the fusion schemes should be able to overcome the limitations of current sensor data. Maintaining the synchronization of data, not only between the digital and physical representations but also between the sensor readings from different sensors, is crucial to maintaining the proper operation of a DT system. However, this is a challenging task due to the changing environment within the physical world and communication constraints [84].

2) *Digital Sibling*: This section presents research challenges related to different components of digital sibling.

a) *Storage*: The TDT architecture requires the storing of a significant amount of data involving complex, expensive storage equipment. Because real-time data will be pouring into and out of the database, the writing and reading speeds of the data storage are just as important as the amount of storage. Real-time data flows into the digital shadow, and historical data is stored in the digital sibling, where time-critical mobility and safety-related applications run in real-time using the digital shadow data. Structured, semi-structured, and unstructured data are generated by mobile and fixed nodes. As a result, the data has Big Data characteristics, i.e., volume, velocity, and variety while also being noisy and heterogeneous. Significant data cleaning operations need to be performed on the collected data in order to make it suitable for processing and storage in databases [111].

b) *Modeling*: The complex physical systems observed in the physical world are simulated via modeling. Two approaches can be used to model a transportation network: 1) physics-based modeling or 2) data-driven modeling [52]. The former type of model relies on physics-based equations

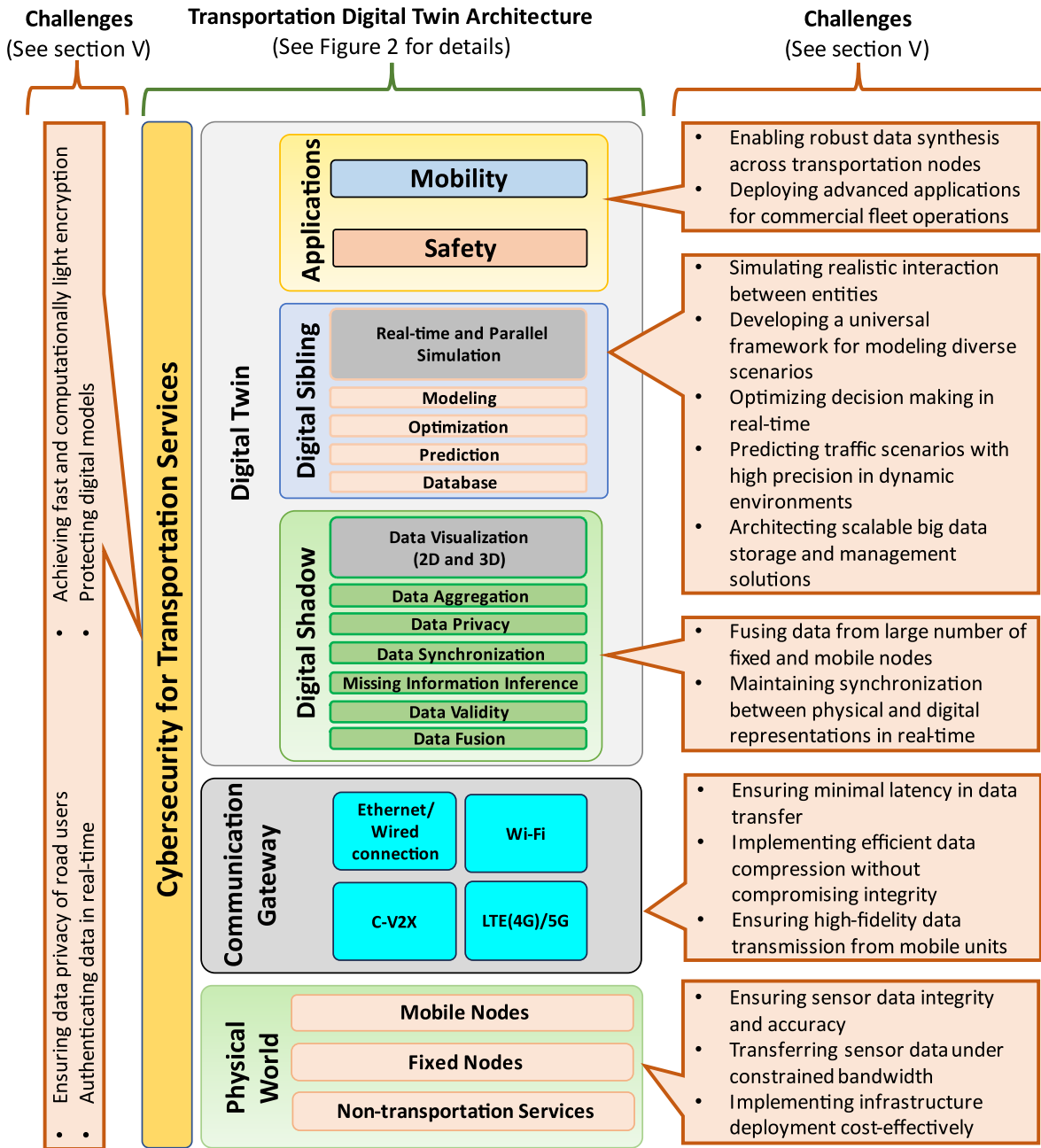


Fig. 3. Research challenges for the TDT architecture.

to model a process or system under certain assumptions about the system. An instance of physics-based model utilization can be found in [43], where multiple physics-based models of a vehicle subsystem were combined to develop the DT model of a braking system to investigate safety. The advantage of such models is that it is easier to interpret the relationship between model parameters and inputs. However, physics-based modeling in transportation system is difficult due to the sheer number of external influences on the system which cannot be accounted for in the model. The underlying assumptions of the model also introduce inaccuracies in the model. Hence, they are not as popular within TDT applications as compared to data-driven models.

As the TDT platform ensures the availability of a massive amount of data, data-driven models are more favored. Data-driven models analyze large volumes of data from a system to establish mathematical relationship between the inputs, outputs and parameters. Since these models are fitted to the real data from the system, they often produce a more accurate relationship between the inputs and outputs and overcome the need to make assumptions as in the case of physics-based models. Thus, in applications, such as short-term traffic flow prediction [92], and adaptive traffic control [91] and data-driven models are ideal. However, data quality and the lack of a balanced data set can significantly reduce the efficacy of data-driven models. Regardless of the type of modeling,

there is a great void of research related to the modeling of the various components within TDT systems [114]. Data interoperability is a challenge pertaining to the formatting of data from various sources to ensure seamless combined operation between systems that are not directly related, for example, traffic flow data and weather data. Another data issue faced in the modeling aspects is the uncertainty in the data [115]. Uncertainties in the data from sensors will get passed to the models utilized in the DTs. Thus, these uncertainties will contaminate the prediction outputs of the models and corrupt the outputs of any models or applications that use the prediction results, thereby causing significant propagation of errors. Furthermore, as real-world transportation data is often fraught with gaps in data, DTs need to address this through the imputation of missing data [65]. Also, there is a lack of a common framework for modeling different entities [116].

c) Simulation: Accurate demand modeling, accurate driving behavior mimicking, and the stochastic nature of traffic are major research challenges for successful traffic simulation. Traditionally, microscopic traffic simulators, such as SUMO, Simcenter Prescan, and PTV Vissim, are used to simulate various transportation scenarios [117], [118]. Moreover, TDT systems may also be used to monitor and predict environmental impacts of the traffic flow in real-time. The microscopic traffic simulators can be coupled with emission simulators, such as motor vehicle emission simulator (MOVES), in order to track pollutants and various form of automotive emissions. By incorporating this as a co-simulation within the DT representation of the transportation system, a real-time emission monitoring system can be implemented within the TDT system. This was demonstrated in [119] for an urban scale DT of the Cebu Airport area combining VISSIM and MOVES simulations. Additionally, Xu et al. [120] combined a virtual reality environment built using Unity and Blender modeling software to monitor emissions in real-time and assess the efficacy of vehicle routing plans within a DT of a real-world urban area. For a large network, the simulation becomes computationally expensive, and they are not suitable for real-time safety-related applications due to the high-computational time requirements. Simulating an entire road network while meeting critical time constraints is a difficult task. Another challenging aspect of the transportation system is the sheer number of interacting entities. These include interactions between vehicles on the roadway as well as those between vehicles and pedestrians. Accurately simulating all those entities would involve running concurrent parallel simulations to test for each variable and each scenario. Moreover, the simulation must also incorporate real-time data that is being fed into the DT. Such large-scale and parallel simulations will require enormous computation power which far exceeds the parallel computing capability of current classical computers [121].

d) Prediction: The ability to perform predictive analysis will be a major value proposition of TDT systems. ML models are mostly used for prediction, which is a data-driven approach. Consequently, if the ML models encounter a data point that significantly differs from the training data set, the algorithms may output inaccurate results. However, most ML models are unable to explain their predictions with respect to

their input data. This prevents the use of such ML models for safety applications. Therefore, challenges exist in this domain that are mostly centered around model training, model interpretability and quality of training data.

e) Optimization: Mobility and safety-related DT applications could be designed to make the best use of both data storage and computational resources. This is more difficult to achieve because the optimization must be done in real-time. Furthermore, local applications optimize traffic, transportation assets, and special event (e.g., traffic incident and gameday) optimization tasks. Network-wide traffic flow optimization is a computationally heavy task and doing so in real time is even more difficult. Once again, quantum computers can have promising applications in this area. Quantum computers could enable the optimization of highly complex traffic optimization problems [122]. Existing traffic optimization algorithms may need to be modified to run on quantum computing systems. Examples of optimization problems that are being investigated using quantum computing are traffic signal optimization [122], traffic flow optimization [123] and shuttle service optimization [124].

C. Communication

As the connecting layer between the physical and cyberspace layers, the communication layer has to deal with all the data exchange requirements of a DT system. With ever-increasing volumes of data being generated by today's sensors and devices, it is a challenge for communication protocols to keep up with the requirements. DT systems are highly reliant upon low-latency communication mediums to ensure seamless synchronization between the digital and the physical world. High communication latency would put them out of sync and defeat the purpose of the DT system [125]. Using high-speed wired connections and low-latency wireless connections like 5G can be a step toward overcoming the latency challenge. Specifically, although 5G could ensure communication latency under 5 ms, TDT systems could involve mobile vehicular networks, and the next generation of wireless technology needs to satisfy them. 6G technology promises below 1-ms latency in communication. One avenue of research is intelligent reflecting surfaces (IRSs), which are intelligent surfaces that can be controlled to direct the reflection of electromagnetic waves from them [96]. This enables them to direct radio waves as required, thereby enhancing signal propagation. In a TDT system, IRS can help to facilitate better communication between various mobile elements in the transportation system.

In order to reduce the amount of data that is required to be transmitted to the cloud, data compression techniques are being explored. Data compression is a challenge due to the various kinds of data that are generated by different types of sensors that will be used in TDT systems. Furthermore, the messaging protocols that are used to transmit data must be lightweight to ensure minimum latency and high-data rates [125], [126]. These challenges arise from the reliance on single modes of communication options and may be addressed by the use of HetNet.

HetNets employ different communication options, such as 5G, LTE, and Wi-Fi with seamless switchover between the various options as required. Data transmission from mobile nodes within the proposed architecture is prone to data packet loss as well as loss of network coverage. Therefore, HetNets can help to resolve these issues and maintain seamless low-latency communication between the physical assets and their digital counterparts and should be investigated for use in TDT systems. Additionally, it must be ensured that during switchovers between the communication options, no security breach occurs. Hence, secure and efficient switchover algorithms need to be developed in future research work to make HetNets viable for use in TDT [127].

D. Privacy and Cybersecurity

Privacy and cybersecurity are at the forefront of systems like TDT which consist of a vast interconnected network that collects data from all entities in a transportation system. DT systems face the problem of protecting digital models and assets within the digital world. If malicious actors are able to gain access to these components, they can explore the attack surface within the DT system and formulate attack models with ease [128]. Data that is collected from vehicles and individuals could potentially be used to identify them. Therefore, any data breach would also entail concerns regarding the privacy of people's data. Storage of such data within databases would require protecting the sensitive information of road users. Hence, anonymity of users needs to be addressed within DT systems. Although data encryption methods exist, they are insufficient to meet the constraints of time and computing resource availability within vehicles [129]. Another related challenge is that of data authentication from mobile nodes, i.e., the identity of the mobile node must be validated before feeding to the DT, to avoid malicious data attacks [129], [130]. However, real-time authentication of the enormous number of fixed and mobile nodes within the TDT architecture is not trivial and remains an open challenge for TDT research. The use of blockchain technology is being investigated to ensure authentication and data integrity within TDT systems [131], [132]. It can address the challenge by providing a secure and transparent ledger that records all the transactions and changes made to the DT. Due to the decentralized nature of the blockchain, it is nearly impossible for one person or entity to change or alter the data because it is stored across a network of computers [89], [131], [133]. Fast, robust and computationally light encryption needs to be developed for use across the computation units found in vehicles.

E. Applications

The TDT applications as reviewed within this article primarily focus on improving safety and mobility of transportation systems. In addition to these, other application categories are possible and have been envisioned in the USDOT ARC-IT for implementation in ITS [1]. These include Commercial Vehicle Operations and weather-based applications amongst others. As mentioned previously, a TDT system may be built

as a hierarchical structure based on whether it is an individual asset twin, a network subcomponent twin or a network level twin. The requirements of a TDT-based application can differ depending on which level of the hierarchy the system is operating in. Apart from the varying requirements of storage, and computation power for data analytics between the levels of the hierarchy, two other aspects of design are the spatial and temporal context of an application. This has already been considered in ARC-IT, where data flows are categorized under 5 categories in the spatial context depending on the distance between the data exchanging elements and 4 categories for the temporal context depending on the how recent the data needs to be for the particular application. The same contexts can be ported over to TDT-based applications and represent important challenges for implementation. Thus, a real-time network level TDT-based mobility application may be desirable, but considering the size and dimensions of the network, challenges may arise if a low-latency communication infrastructure is not available over the entire network for data flow to occur. It may require the integration of data from numerous individual level twins or network subcomponent level twins whilst satisfying temporal context requirement of the application. Thus, aggregating data across time and space is a major challenge in developing the applications.

VI. FUTURE RESEARCH DIRECTIONS

This section condenses the various gaps identified from within the literature review and presents our perspective for future research directions for deployment ready TDT systems. These perspectives will help to direct future research direction in this realm.

A. Multiaccess Edge Computing

At the physical world level, due to the requirement of computation power and low-latency communication of processing results, a cloud architecture may be inadequate for the system. Furthermore, exchanging the collected physical world data to a remote server will induce unnecessary network congestion and poor efficiency. To enable faster computation at the edge level, an edge computing node, such as MEC node, can be used. MEC servers provide accessible computing power to mobile elements, such as vehicles, thereby enabling the offloading of computing tasks to the edge [134]. As they are in close proximity to the actual physical assets, communication latency can be ensured at the minimum possible level. The integration of DTs with MEC is known as DT edge network (DITEN), which facilitates the real time execution of applications depending on the computational and communication latency requirements. Research into DITENs has focused on intelligent task offloading schemes for autonomous vehicles [98], [135]. In addition, in order to provide computing services whilst ensuring low latency, the location of the MEC services should be optimized to ensure they are near to the physical world assets. Thus, location optimization of MEC services is an open area of research.

B. Quantum Sensing

An avenue of future research is quantum sensors, which are sensors that measure physical parameters, including magnetic fields, electric fields, temperature, and pressure with great precision, using the principles of quantum mechanics. Most noteworthy is that they can surpass the capabilities of classical sensor technologies [136]. Recently, quantum sensors have shown great promise in navigation applications in situations where satellite-based positioning systems may not be available [137]. Similarly, quantum-enhanced LiDARs that have greater range and sensitivity are also being developed [138], [139]. Thus, they can essentially remove the effect of environmental occlusions on sensors. However, these sensors are currently too large and expensive to be widely deployed. Further research will need to be conducted to make them viable in terms of cost and dimension before they can be utilized for DT applications. DQS is a new paradigm that can help to improve the performance of a network of multiple sensors as is found within DT-based systems [136]. Applications to improve sensing, such as in navigation tasks, have already been proposed using a DQS-based approach [140].

C. Data Storage and Processing

Data collection and processing in a networking context is challenging and expensive and requires using common telemetry systems to gather relevant network-related data. Storage of short network data flows can exceed several hundred gigabytes, making it difficult to store and process [141]. Hence, for facilitating data aggregation, research should be conducted into data compression techniques. Extensive data validation needs to be performed before data can be fed into models. Any unchecked inconsistencies or errors in the data will propagate throughout the DT and result in incorrect actuation commands being sent out to the physical world. The challenge here is performing the data validation for a large number of sensors in a large-scale TDT system and executing the validation tasks within a sufficiently small computation time to not cause unacceptable delays in the entire system. While it is ideally assumed that the sensors in a TDT system will be operational with a 100 percent up time, in reality, massive sensor deployments are often fraught with significant amounts of downtime or disruptions. In such instances, the models within the DT may encounter the problem of incomplete or missing information. In a data-driven paradigm, such as DT, such lapses in information can cause interruptions in system operations. Therefore, it is necessary to design missing information inference systems to counteract such temporary disruptions in data and information to ensure seamless system operation.

Future research should focus on building storage facilities that can satisfy the needs of massive TDT systems. Additionally, future data storage needs to be able to serve the millions of data write and data query operations that will be generated by all of the physical components of the TDT system. Fast compression algorithms need to be developed in order to ensure data can be quickly compressed and stored.

The development of database systems specifically tailored for use in TDT systems can greatly help to reduce data writing and reading speeds.

D. Modeling

Future transportation research should aim to develop models for the transportation elements of a TDT system. Because of the high level of uncertainty in human behavior, accurately modeling the physical world is extremely challenging. Extensive research is required to model the behavior of ORUs in a roadway environment. In addition to model development, extensive validation of the developed models needs to be performed in order to ensure applicability. In order to build systems suitable for use in safety-critical applications, future research in ML models should be toward explainable models. Another helpful paradigm is PIML which achieves greater performance by incorporating elements from physics-based modeling into the ML model [142].

GANs are a type of ANN learning architecture that can learn to produce artificial data samples that are remarkably similar to actual data. GAN-based systems can be used to generate synthetic data for training models. Additionally, the incorporation of GAN-based models into DTs have also been explored in other domain [142], [143], and future research on GAN-based modeling in transportation can certainly benefit DTs-based on transportation systems.

E. Quantum Computing

In this regard, a promising future alternative that is currently in development is quantum computing. Quantum computers will have the advantage of being able to run highly complex parallel operations. This will empower the TDT system to analyze what-if situations and precisely determine the outcome of any decision on the transportation network [144]. Researchers have already sought to explore transportation problems using a hybrid quantum-classical approach [145]. Currently, cloud-based simulators enable users to run and test various quantum algorithms for a small number of qubits [146]. However, as they are still in their infancy, quantum computers are now constrained in the amount of qubits they can reliably handle. Large-scale traffic simulations using quantum computers are challenging due to this. As viable quantum computers become a reality in the near future, it is necessary to start developing algorithms that can run traffic simulations efficiently on a quantum computing platform.

F. Applications

As mentioned in the applications challenges section of section V, even though a number of application categories have been envisioned by the USDOT for ITS, current TDT-based applications mainly focus on safety and mobility-based applications. Future research on TDT applications should explore other areas as well, such as spot weather impact warning [1]. Additionally, the reviewed research articles are concentrated toward lower levels (i.e., individual asset twin and network subcomponents twin) of the presented hierarchy as shown in Fig. 1(b). These systems will not suffice for network

level implementations of the same applications as scaling them would impose additional requirements and constraints (spatial and temporal). Future research should be directed toward building distributed and scalable TDT systems and applications given the scale of transportation networks [147]. Increasing autonomy within vehicles has been the prevalent trend in recent years. Therefore, any applications developed for TDT systems should account for vehicle automation capabilities and support them in order to achieve safe and efficient autonomous mobility in the future.

VII. CONCLUSION

With the ever-growing interest in DT systems across different disciplines, the field of transportation systems will also experience widespread adoption of this technology in the near future. This article begins by introducing the concept of TDT and related aspects. Then it discusses the relationship of DT with transportation applications using the context of the ARC-IT framework and associated service packages. This is followed by a hierarchical concept of a TDT system with due consideration to the required complexity, breadth, and depth of the system. Additionally, a reference architecture is presented for a TDT system which is developed from existing literature in transportation and other domains. An in-depth discussion related to each part of the architecture is provided. We analyzed the existing research studies in this field related to safety and mobility and presented a comprehensive review of the current status quo of research in TDT systems. In conjunction with the reference architecture, the hierarchical concept could enable transportation planners to design TDT systems by satisfying corresponding system requirements. Despite the current advances in research, several challenges exist that impede the development of current ITS toward TDT systems. These challenges range from communication requirements to cybersecurity, data storage and modeling limitations. Thus, we investigated these challenges and reviewed them and presented future perspectives with regards TDT research. Therefore, the review is anticipated to provide a holistic and up-to-date overview of TDT systems and will benefit practitioners and future researchers to design TDT systems suited to any safety and mobility needs.

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REFERENCES

- [1] T. Lusco, *ARC-IT—The Architecture Reference for Cooperative and Intelligent Transportation*, U.S. Dept Transp., Washington, DC, USA, Apr. 2018.
- [2] J. Barbaresso et al., *“USDOT’s Intelligent Transportation Systems (ITS) ITS Strategic Plan, 2015-2019*, U.S. Dept. Transp., Washington, DC, USA, 2014.
- [3] W. A. Ali, M. Roccotelli, and M. P. Fantì, “Digital twin in intelligent transportation systems: A review,” in *Proc. 8th Int. Conf. Control, Decision Inf. Technol.*, 2022, pp. 576–581, doi: [10.1109/CoDIT55151.2022.9804017](https://doi.org/10.1109/CoDIT55151.2022.9804017).
- [4] L. Bao, Q. Wang, and Y. Jiang, “Review of digital twin for intelligent transportation system,” in *Proc. Int. Conf. Inf. Control, Electr. Eng. Rail Transit*, 2021, pp. 309–315, doi: [10.1109/ICEERT53919.2021.00064](https://doi.org/10.1109/ICEERT53919.2021.00064).
- [5] M. Kosacka-Olejnik, M. Kostrzewski, M. Marczevska, B. Mrówczyńska, and P. Pawlewski, “How digital twin concept supports internal transport systems?—Literature review,” *Energies*, vol. 14, no. 16, p. 4919, Aug. 2021, doi: [10.3390/en14164919](https://doi.org/10.3390/en14164919).
- [6] C. Schwarz and Z. Wang, “The role of digital twins in connected and automated vehicles,” *IEEE Intell. Transp. Syst. Mag.*, vol. 14, no. 6, pp. 41–51, Nov./Dec. 2022, doi: [10.1109/MITS.2021.3129524](https://doi.org/10.1109/MITS.2021.3129524).
- [7] K. Shafique, B. A. Khawaja, F. Sabir, S. Qazi, and M. Mustaqim, “Internet of Things (IoT) for next-generation smart systems: A review of current challenges, future trends and prospects for emerging 5G-IoT scenarios,” *IEEE Access*, vol. 8, pp. 23022–23040, 2020, doi: [10.1109/ACCESS.2020.2970118](https://doi.org/10.1109/ACCESS.2020.2970118).
- [8] M. Grieves, “Digital twin: manufacturing excellence through virtual factory replication,” Dassault Systèmes, Vélizy-Villacoublay, France, White Paper, 2014.
- [9] Q. Qi and F. Tao, “Digital twin and big data towards smart manufacturing and industry 4.0: 360 degree comparison,” *IEEE Access*, vol. 6, pp. 3585–3593, 2018, doi: [10.1109/ACCESS.2018.2793265](https://doi.org/10.1109/ACCESS.2018.2793265).
- [10] M. Liu, S. Fang, H. Dong, and C. Xu, “Review of digital twin about concepts, technologies, and industrial applications,” *J. Manuf. Syst.*, vol. 58, pp. 346–361, Jan. 2021, doi: [10.1016/j.jmsys.2020.06.017](https://doi.org/10.1016/j.jmsys.2020.06.017).
- [11] W. Kritzing, M. Karner, G. Traar, J. Henjes, and W. Sihm, “Digital Twin in manufacturing: A categorical literature review and classification,” *IFAC-PapersOnLine*, vol. 51, no. 11, pp. 1016–1022, Jan. 2018, doi: [10.1016/j.ifacol.2018.08.474](https://doi.org/10.1016/j.ifacol.2018.08.474).
- [12] Y. Lu, C. Liu, I. Kevin, K. Wang, H. Huang, and X. Xu, “Digital Twin-driven smart manufacturing: Connotation, reference model, applications and research issues,” *Robot Comput. Integr. Manuf.*, vol. 61, Feb. 2020, Art. no. 101837.
- [13] Y. Zhang, C. Zhang, J. Yan, C. Yang, and Z. Liu, “Rapid construction method of equipment model for discrete manufacturing digital twin workshop system,” *Robot Comput. Integr. Manuf.*, vol. 75, Jun. 2022, Art. no. 102309.
- [14] B. He and K.-J. Bai, “Digital twin-based sustainable intelligent manufacturing: A review,” *Adv. Manuf.*, vol. 9, pp. 1–21, Mar. 2021.
- [15] E. Sycheva and P. Shpak, “Practical application of the concept of digital twins in the aviation sector,” in *Digital Transformation in Industry: Digital Twins and New Business Models*. Cham, Switzerland: Springer, 2022, pp. 29–42.
- [16] M. Xiong and H. Wang, “Digital twin applications in aviation industry: A review,” *Int. J. Adv. Manuf. Technol.*, vol. 121, no. 9–10, pp. 5677–5692, 2022.
- [17] J. Liu et al., “Digital twin civil aviation research airport for aircraft security and environment protection,” in *Proc. IEEE 4th Int. Conf. Civil Aviat. Saf. Inf. Technol. (ICCASIT)*, 2022, pp. 408–412.
- [18] H. Pan, Z. Dou, Y. Cai, W. Li, X. Lei, and D. Han, “Digital twin and its application in power system,” in *Proc. 5th Int. Conf. Power Renew. Energy*, 2020, pp. 21–26, doi: [10.1109/ICPRE51194.2020.9233278](https://doi.org/10.1109/ICPRE51194.2020.9233278).
- [19] Z. Jiang, H. Lv, Y. Li, and Y. Guo, “A novel application architecture of digital twin in smart grid,” *J. Ambient Intell. Humaniz. Comput.*, vol. 13, no. 8, pp. 3819–3835, 2022.
- [20] M. Atalay and P. Angin, “A digital twins approach to smart grid security testing and standardization,” in *Proc. IEEE Int. Workshop Metrol. Ind. IoT*, 2020, pp. 435–440.
- [21] A. K. Sleiti, J. S. Kapat, and L. Vesely, “Digital twin in energy industry: Proposed robust digital twin for power plant and other complex capital-intensive large engineering systems,” *Energy Rep.*, vol. 8, pp. 3704–3726, Nov. 2022.
- [22] M. Jahanshahi Zeitouni, A. Parvaresh, S. Abrazeh, S.-R. Mohseni, M. Gheisarnejad, and M.-H. Khooban, “Digital twins-assisted design of next-generation advanced controllers for power systems and electronics: Wind turbine as a case study,” *Inventions*, vol. 5, no. 2, p. 19, 2020.
- [23] P. Palensky, M. Cvetkovic, D. Gusain, and A. Joseph, “Digital twins and their use in future power systems,” *Digit. Twin*, vol. 1, no. 4, p. 4, 2022.
- [24] X. Liao et al., “Cooperative ramp merging design and field implementation: A digital twin approach based on vehicle-to-cloud communication,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 5, pp. 4490–4500, May 2022, doi: [10.1109/TITS.2020.3045123](https://doi.org/10.1109/TITS.2020.3045123).
- [25] Z. Wang, K. Han, and P. Tiwari, “Digital twin-assisted cooperative driving at non-signalized intersections,” *IEEE Trans. Intell. Veh.*, vol. 7, no. 2, pp. 198–209, Jun. 2022, doi: [10.1109/TIV.2021.3100465](https://doi.org/10.1109/TIV.2021.3100465).

- [26] X. Chen, E. Kang, S. Shiraishi, V. M. Preciado, and Z. Jiang, "Digital behavioral twins for safe connected cars," in *Proc. 21st ACM/IEEE Int. Conf. Model Driven Eng. Lang. Syst.*, 2018, pp. 144–154, doi: [10.1145/3239372.3239401](https://doi.org/10.1145/3239372.3239401).
- [27] S. A. P. Kumar, R. Madhumathi, P. R. Chelliah, L. Tao, and S. Wang, "A novel digital twin-centric approach for driver intention prediction and traffic congestion avoidance," *J. Rel. Intell. Environ.*, vol. 4, no. 4, pp. 199–209, Dec. 2018, doi: [10.1007/s40860-018-0069-y](https://doi.org/10.1007/s40860-018-0069-y).
- [28] Z. Wang et al., "A digital twin paradigm: Vehicle-to-cloud based advanced driver assistance systems," in *Proc. IEEE 91st Veh. Technol. Conf.*, 2020, pp. 1–6.
- [29] A. J. Saroj, S. Roy, A. Guin, and M. Hunter, "Development of a connected corridor real-time data-driven traffic digital twin simulation model," *J. Transp. Eng. A, Syst.*, vol. 147, no. 12, Dec. 2021, doi: [10.1061/jtepbs.0000599](https://doi.org/10.1061/jtepbs.0000599).
- [30] A. Saroj, T. V. Trant, A. Guin, M. Hunter, and M. Sartipi, "Optimizing traffic controllers along the MLK smart corridor using reinforcement learning and digital twin," in *Proc. IEEE 2nd Int. Conf. Digital Twins Parallel Intell.*, 2022, pp. 1–2, doi: [10.1109/DTP155838.2022.9998928](https://doi.org/10.1109/DTP155838.2022.9998928).
- [31] H. Banaee, F. Klügl, and A. Loutfi, "Challenges in using digital twins for modelling human behaviour in environments," in *Proc. Neuro-Cogn. Model. Hum. Environ.*, 2021, pp. 1–3. [Online]. Available: <https://www.researchgate.net/publication/344298318>
- [32] Z. Fan et al., "Online trajectory prediction for metropolitan scale mobility digital twin," in *Proc. ACM Int. Symp. Adv. Geograph. Inf. Syst.*, 2022, pp. 1–12, doi: [10.1145/3557915.3561040](https://doi.org/10.1145/3557915.3561040).
- [33] S. K. Sowe, E. Simmon, K. Zettsu, F. De Vault, and I. Bojanova, "Cyber-physical-human systems: Putting people in the loop," *IT Prof.*, vol. 18, no. 1, pp. 10–13, Jan. 2016, doi: [10.1109/MITP.2016.14](https://doi.org/10.1109/MITP.2016.14).
- [34] D. Sahinel, C. Akpolat, O. C. Görür, and F. Sivrikaya, "Integration of human actors in IoT and CPS landscape," in *Proc. IEEE 5th World Forum Internet Things*, 2019, pp. 485–490.
- [35] *Untangling the Requirements of a Digital Twin*, Univ. Sheffield, Sheffield, U.K., Oct. 2020.
- [36] G. Pronost, F. Mayer, B. Marche, M. Camargo, and L. Dupont, "Towards a framework for the classification of digital twins and their applications," in *Proc. IEEE Int. Conf. Eng., Technol. Innov.*, 2021, pp. 1–7, doi: [10.1109/ICE/ITMC52061.2021.9570114](https://doi.org/10.1109/ICE/ITMC52061.2021.9570114).
- [37] D. Jones, C. Snider, A. Nassehi, J. Yon, and B. Hicks, "Characterising the digital twin: A systematic literature review," *CIRP J. Manuf. Sci. Technol.*, vol. 29, pp. 36–52, May 2020, doi: [10.1016/j.cirpj.2020.02.002](https://doi.org/10.1016/j.cirpj.2020.02.002).
- [38] M. Grieves and J. Vickers, "Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems," in *Transdisciplinary Perspectives on Complex Systems: New Findings and Approaches*. Cham, Switzerland: Springer, 2017, pp. 85–113, doi: [10.1007/978-3-319-38756-7_4](https://doi.org/10.1007/978-3-319-38756-7_4).
- [39] I. Barosan, A. A. Basmenj, S. G. R. Chouhan, and D. Manrique, "Development of a virtual simulation environment and a digital twin of an autonomous driving truck for a distribution center," in *Proc. Eur. Conf. Softw. Archit.*, 2020, pp. 542–557, doi: [10.1007/978-3-030-59155-7_39](https://doi.org/10.1007/978-3-030-59155-7_39).
- [40] A. Krämer et al., "Providentia—A large-scale sensor system for the assistance of autonomous vehicles and its evaluation," 2019, *arXiv:1906.06789*.
- [41] O. el Marai, T. Taleb, and J. Song, "Roads infrastructure digital twin: A step toward smarter cities realization," *IEEE Netw.*, vol. 35, no. 2, pp. 136–143, Mar. 2021, doi: [10.1109/MNET.011.2000398](https://doi.org/10.1109/MNET.011.2000398).
- [42] K. M. Alam and A. El Saddik, "C2PS: A digital twin architecture reference model for the cloud-based cyber-physical systems," *IEEE Access*, vol. 5, pp. 2050–2062, 2017, doi: [10.1109/ACCESS.2017.2657006](https://doi.org/10.1109/ACCESS.2017.2657006).
- [43] V. Dygalo, A. Keller, and A. Shcherbin, "Principles of application of virtual and physical simulation technology in production of digital twin of active vehicle safety systems," *Transp. Res. Procedia*, 2020, pp. 121–129, doi: [10.1016/j.trpro.2020.10.015](https://doi.org/10.1016/j.trpro.2020.10.015).
- [44] Z. Wang, K. Han, and P. Tiwari, "Digital twin simulation of connected and automated vehicles with the unity game engine," in *Proc. IEEE 1st Int. Conf. Digit. Twins Parallel Intell.*, 2021, pp. 180–183, doi: [10.1109/DTP152967.2021.9540074](https://doi.org/10.1109/DTP152967.2021.9540074).
- [45] S. Almeida, S. Al-Rubaye, A. Tsourdos, and N. P. Avdelidis, "Digital twin analysis to promote safety and security in autonomous vehicles," *IEEE Commun. Stand. Mag.*, vol. 5, no. 1, pp. 40–46, Mar. 2021, doi: [10.1109/MCOMSTD.011.2100004](https://doi.org/10.1109/MCOMSTD.011.2100004).
- [46] Y. Ge, Y. Wang, R. Yu, Q. Han, and Y. Chen, "Research on test method of autonomous driving based on digital twin," in *Proc. IEEE Veh. Netw. Conf. (VNC)*, 2019, pp. 1–2.
- [47] Z. Liu, N. Meyendorf, and N. Mrad, "The role of data fusion in predictive maintenance using digital twin," *AIP Conf. Proc.*, vol. 1949, no. 1, 2018, Art. no. 20023, doi: [10.1063/1.5031520](https://doi.org/10.1063/1.5031520).
- [48] A. Rudskoy, I. Ilin, and A. Prokhorov, "Digital twins in the intelligent transport systems," *Transp. Res. Procedia*, vol. 54, pp. 927–935, Jan. 2021, doi: [10.1016/j.trpro.2021.02.152](https://doi.org/10.1016/j.trpro.2021.02.152).
- [49] M. Schellmann, M. Buechel, H. Rosier, T. Kessler, and A. Knoll, "Fortuna: Presenting the 5G-connected automated vehicle prototype of the project PROVIDENTIA," 2019, Preprint. [Online]. Available: <http://dx.doi.org/10.13140/RG.2.2.24402.91842>
- [50] A. Stanford-Clark, E. Frank-Schultz, and M. Harris, "What are digital twins?" Accessed: Oct. 11, 2022. [Online]. Available: <https://developer.ibm.com/articles/what-are-digital-twins/>
- [51] Z. Wang et al., "Mobility digital twin: Concept, architecture, case study, and future challenges," *IEEE Internet Things J.*, vol. 9, no. 18, pp. 17452–17467, Sep. 2022, doi: [10.1109/JIOT.2022.3156028](https://doi.org/10.1109/JIOT.2022.3156028).
- [52] A. Rasheed, O. San, and T. Kvamsdal, "Digital twin: Values, challenges and enablers from a modeling perspective," *IEEE Access*, vol. 8, pp. 21980–22012, 2020, doi: [10.1109/ACCESS.2020.2970143](https://doi.org/10.1109/ACCESS.2020.2970143).
- [53] J. Kocic, N. Jovicic, and V. Drndarevic, "Sensors and sensor fusion in autonomous vehicles," in *Proc. 26th Telecommun. Forum (TELFOR)*, 2018, pp. 420–425, doi: [10.1109/TELFOR.2018.8612054](https://doi.org/10.1109/TELFOR.2018.8612054).
- [54] K. Merry and P. Bettinger, "Smartphone GPS accuracy study in an urban environment," *PLoS One*, vol. 14, no. 7, Jul. 2019, Art. no. e0219890, doi: [10.1371/journal.pone.0219890](https://doi.org/10.1371/journal.pone.0219890).
- [55] J. Zhao, H. Xu, H. Liu, J. Wu, Y. Zheng, and D. Wu, "Detection and tracking of pedestrians and vehicles using roadside LiDAR sensors," *Transp. Res. Part C, Emerg. Technol.*, vol. 100, pp. 68–87, Mar. 2019, doi: [10.1016/j.trc.2019.01.007](https://doi.org/10.1016/j.trc.2019.01.007).
- [56] B. Qi, W. Zhao, H. Zhang, Z. Jin, X. Wang, and T. Runge, "Automated traffic volume analytics at road intersections using computer vision techniques," in *Proc. 5th Int. Conf. Transp. Inf. Saf. (ICTIS)*, 2019, pp. 161–169.
- [57] M. M. Hamdi, L. Audah, S. A. Rashid, and M. Al Shareeda, "Techniques of early incident detection and traffic monitoring centre in VANETs: A review," *J. Commun.*, vol. 15, no. 12, pp. 896–904, 2020.
- [58] H.-S. Lim, H.-M. Park, J.-E. Lee, Y.-H. Kim, and S. Lee, "Lane-by-lane traffic monitoring using 24.1 GHz FMCW radar system," *IEEE Access*, vol. 9, pp. 14677–14687, 2021.
- [59] L. Cao, T. Wang, D. Wang, K. Du, Y. Liu, and C. Fu, "Lane determination of vehicles based on a novel clustering algorithm for intelligent traffic monitoring," *IEEE Access*, vol. 8, pp. 63004–63017, 2020.
- [60] C. He, T. H. Luan, R. Lu, Z. Su, and M. Dong, "Security and privacy in vehicular digital twin networks: Challenges and solutions," *IEEE Wireless Commun.*, vol. 30, no. 4, pp. 154–160, Aug. 2023.
- [61] Y. Wu, K. Zhang, and Y. Zhang, "Digital twin networks: A survey," *IEEE Internet Things J.*, vol. 8, no. 18, pp. 13789–13804, Sep. 2021, doi: [10.1109/JIOT.2021.3079510](https://doi.org/10.1109/JIOT.2021.3079510).
- [62] I. Sourmey, *The Impact of the Communication Technology Protocol on Your IoT Application's Power Consumption*, Saft, Levallois-Perret, France, 2020.
- [63] *Choose the Right IoT Wireless Connectivity Protocol*, Abracon LLC, Austin, TX, USA, 2018.
- [64] M. Singh, E. Fuenmayor, E. Hinchy, Y. Qiao, N. Murray, and D. Devine, "Digital twin: Origin to future," *Appl. Syst. Innov.*, vol. 4, no. 2, p. 36, May 2021, doi: [10.3390/asi4020036](https://doi.org/10.3390/asi4020036).
- [65] A. J. Saroj, S. Roy, A. Guin, and M. Hunter, "Impact of connected corridor volume data imputations on digital twin performance measures," *Int. J. Transp. Sci. Technol.*, vol. 12, no. 2, pp. 476–491, 2023, doi: [10.1016/j.ijtst.2022.05.001](https://doi.org/10.1016/j.ijtst.2022.05.001).
- [66] T. Bergs, S. Gierlings, T. Auerbach, A. Klink, D. Schraknepper, and T. Augspurger, "The concept of digital twin and digital shadow in manufacturing," *Procedia CIRP*, vol. 101, pp. 81–84, Jan. 2021, doi: [10.1016/j.procir.2021.02.010](https://doi.org/10.1016/j.procir.2021.02.010).
- [67] A. Ghose, J. Horkoff, V. E. S. Souza, J. Parsons, and J. Evermann, "Conceptual modeling," in *Proc. Int. Conf. Concept. Model.*, 2021, pp. 271–281, doi: [10.1007/978-3-030-89022-3](https://doi.org/10.1007/978-3-030-89022-3).
- [68] A. AboElHassan and S. Yacout, "A digital shadow framework using distributed system concepts," *J. Intell. Manuf.*, vol. 183, Sep. 2023, Art. no. 109534, doi: [10.1007/s10845-022-02028-6](https://doi.org/10.1007/s10845-022-02028-6).

- [69] A. R. Al-Ali, R. Gupta, T. Z. Batool, T. Landolsi, F. Aloul, and A. Al Nabulsi, "Digital twin conceptual model within the context of Internet of Things," *Future Internet*, vol. 12, no. 10, pp. 1–15, Oct. 2020, doi: [10.3390/fi12100163](https://doi.org/10.3390/fi12100163).
- [70] Y. Zheng, S. Yang, and H. Cheng, "An application framework of digital twin and its case study," *J. Ambient Intell. Humaniz. Comput.*, vol. 10, no. 3, pp. 1141–1153, Mar. 2019, doi: [10.1007/s12652-018-0911-3](https://doi.org/10.1007/s12652-018-0911-3).
- [71] M. Kaliske, R. Behnke, and I. Wollny, "Vision on a digital twin of the road-tire-vehicle system for future mobility," *Tire Sci. Technol.*, vol. 49, no. 1, pp. 2–18, 2021. [Online]. Available: <http://meridian.allenpress.com/ts/article-pdf/49/1/2/2825823/fi1945-5852-49-1-2.pdf>
- [72] A. A. Korostelkin, O. I. Klyavin, M. V. Aleshin, W. Guodong, W. Suifeng, and L. Jingyi, "Optimization of frame mass in crash testing of off-road vehicles," *Russian Eng. Res.*, vol. 39, no. 12, pp. 1021–1028, Dec. 2019, doi: [10.3103/S1068798X19120116](https://doi.org/10.3103/S1068798X19120116).
- [73] L. Nie, X. Wang, Q. Zhao, Z. Shang, L. Feng, and G. Li, "Digital twin for transportation big data: A reinforcement learning-based network traffic prediction approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 1, pp. 896–906, Jan. 2024, doi: [10.1109/tits.2022.3232518](https://doi.org/10.1109/tits.2022.3232518).
- [74] L. Zhao, G. Han, Z. Li, and L. Shu, "Intelligent digital twin-based software-defined vehicular networks," *IEEE Netw.*, vol. 34, no. 5, pp. 178–184, Sep./Oct. 2020, doi: [10.1109/MNET.011.1900587](https://doi.org/10.1109/MNET.011.1900587).
- [75] A. L. C. Bazzan and F. Klügl, *Multi-Agent Systems for Traffic and Transportation Engineering*. Hershey, PA, USA: IGI Global, 2009.
- [76] P. Gruer, V. Hilaire, and A. Koukarn, "Multi-agent approach to modeling and simulation of urban transportation systems," in *Proc. IEEE Int. Conf. Syst., Man Cybern. e-Syst. e-Man Cybern. Cyberspace*, 2001, pp. 2499–2504.
- [77] S. Satunin and E. Babkin, "A multi-agent approach to intelligent transportation systems modeling with combinatorial auctions," *Expert Syst. Appl.*, vol. 41, no. 15, pp. 6622–6633, 2014.
- [78] M. de Bok, L. Tavasszy, and S. Thoen, "Application of an empirical multi-agent model for urban goods transport to analyze impacts of zero emission zones in The Netherlands," *Transp. Policy*, vol. 124, pp. 119–127, Aug. 2022, doi: [10.1016/j.tranpol.2020.07.010](https://doi.org/10.1016/j.tranpol.2020.07.010).
- [79] H. Hamidi and A. Kamankesh, "An approach to intelligent traffic management system using a multi-agent system," *Int. J. Intell. Transp. Syst. Res.*, vol. 16, no. 2, pp. 112–124, May 2018, doi: [10.1007/s13177-017-0142-6](https://doi.org/10.1007/s13177-017-0142-6).
- [80] B. Torabi, M. Al-Zinati, and R. Z. Wenkstern, "Matisse 3.0: A large-scale multi-agent simulation system for intelligent transportation systems," in *Proc. 16th Int. Conf. Adv. Pract. Appl. Agents, Multi-Agent Syst., Complex. PAAMS Collect. (PAAMS)*, 2018, pp. 357–360.
- [81] W. Lepuschitz, B. Groessing, M. Merdan, and G. Schitter, "Evaluation of a multi-agent approach for a real transportation system," in *Proc. IEEE Int. Conf. Ind. Technol. (ICIT)*, 2013, pp. 1273–1278.
- [82] M. Balmer, K. Nagel, and B. Raney, "Large-scale multi-agent simulations for transportation applications," *J. Intell. Transp. Syst.*, vol. 8, no. 4, pp. 205–221, 2004.
- [83] R. Minerva, G. M. Lee, and N. Crespi, "Digital twin in the IoT context: A survey on technical features, scenarios, and architectural models," *Proc. IEEE*, vol. 108, no. 10, pp. 1785–1824, Oct. 2020, doi: [10.1109/JPROC.2020.2998530](https://doi.org/10.1109/JPROC.2020.2998530).
- [84] T. H. Luan, R. Liu, L. Gao, R. Li, and H. Zhou, "The paradigm of digital twin communications," 2021, *arXiv:2105.07182*.
- [85] Q. Guo, F. Tang, T. K. Rodrigues, and N. Kato, "Five disruptive technologies in 6G to support digital twin networks," *IEEE Wireless Commun.*, vol. 31, no. 1, pp. 149–155, Feb. 2024.
- [86] Y. Liu, Z. Wang, K. Han, Z. Shou, P. Tiwari, and J. H. L. Hansen, "Sensor fusion of camera and cloud digital twin information for intelligent vehicles," in *Proc. IEEE Intell. Veh. Symp.*, 2020, pp. 182–187, doi: [10.1109/IV47402.2020.9304643](https://doi.org/10.1109/IV47402.2020.9304643).
- [87] A. Krämer et al., "Providentia—a large-scale sensor system for the assistance of autonomous vehicles and its evaluation," 2021, *arXiv:1906.06789*.
- [88] A. Adibfar and A. M. Costin, "Creation of a mock-up bridge digital twin by fusing intelligent transportation systems (ITS) data into bridge information model (BrIM)," *J. Constr. Eng. Manag.*, vol. 148, no. 9, Sep. 2022, Art. no. 04022094, doi: [10.1061/\(asce\)co.1943-7862.0002332](https://doi.org/10.1061/(asce)co.1943-7862.0002332).
- [89] R. Sahal, S. H. Alsamhi, K. N. Brown, D. O'shea, C. McCarthy, and M. Guizani, "Blockchain-empowered digital twins collaboration: Smart transportation use case," *Machines*, vol. 9, no. 9, p. 193, Sep. 2021, doi: [10.3390/machines9090193](https://doi.org/10.3390/machines9090193).
- [90] D. Piromalis and A. Kantaros, "Digital twins in the automotive industry: The road toward physical-digital convergence," *Appl. Syst. Innov.*, vol. 5, no. 4, p. 65, Aug. 2022, doi: [10.3390/asi5040065](https://doi.org/10.3390/asi5040065).
- [91] S. Dasgupta, M. Rahman, A. D. Lidbe, W. Lu, and S. Jones, "A transportation digital-twin approach for adaptive traffic control systems," 2023, *arXiv:2109.10863*.
- [92] Z. Tu, L. Qiao, R. Nowak, H. Lv, and Z. Lv, "Digital twins-based automated pilot for energy-efficiency assessment of intelligent transportation infrastructure," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22320–22330, Nov. 2022, doi: [10.1109/TITS.2022.3166585](https://doi.org/10.1109/TITS.2022.3166585).
- [93] H. Xu et al., "Smart mobility in the cloud: enabling real-time situational awareness and cyber-physical control through a digital twin for traffic," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 3, pp. 3145–3156, Mar. 2023, doi: [10.1109/TITS.2022.3226746](https://doi.org/10.1109/TITS.2022.3226746).
- [94] Y. Li and W. Zhang, "Traffic flow digital twin generation for highway scenario based on radar-camera paired fusion," *Sci. Rep.*, vol. 13, no. 1, p. 642, Dec. 2023, doi: [10.1038/s41598-023-27696-z](https://doi.org/10.1038/s41598-023-27696-z).
- [95] T. Zhang, X. Liu, Z. Luo, F. Dong, and Y. Jiang, "Time series behavior modeling with digital twin for Internet of Vehicles," *EURASIP J. Wireless Commun. Netw.*, vol. 2019, no. 1, p. 271, Dec. 2019, doi: [10.1186/s13638-019-1589-8](https://doi.org/10.1186/s13638-019-1589-8).
- [96] X. Yuan, J. Chen, N. Zhang, J. Ni, F. R. Yu, and V. C. M. Leung, "Digital twin-driven vehicular task offloading and IRS configuration in the Internet of Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 24290–24304, Dec. 2022, doi: [10.1109/TITS.2022.3204585](https://doi.org/10.1109/TITS.2022.3204585).
- [97] K. Kušić, R. Schumann, and E. Ivanjko, "A digital twin in transportation: Real-time synergy of traffic data streams and simulation for virtualizing motorway dynamics," *Adv. Eng. Inform.*, vol. 55, Jan. 2023, Art. no. 101858, doi: [10.1016/j.aei.2022.101858](https://doi.org/10.1016/j.aei.2022.101858).
- [98] X. Xu et al., "Service offloading with deep Q-network for digital twinning-empowered Internet of Vehicles in edge computing," *IEEE Trans. Industr. Inform.*, vol. 18, no. 2, pp. 1414–1423, Feb. 2022, doi: [10.1109/TII.2020.3040180](https://doi.org/10.1109/TII.2020.3040180).
- [99] C. Hu et al., "Digital twin-assisted real-time traffic data prediction method for 5G-enabled Internet of Vehicles," *IEEE Trans. Industr. Inform.*, vol. 18, no. 4, pp. 2811–2819, Apr. 2022, doi: [10.1109/TII.2021.3083596](https://doi.org/10.1109/TII.2021.3083596).
- [100] H. Feng, H. Lv, and Z. Lv, "Resilience towarded Digital Twins to improve the adaptability of transportation systems," *Transp. Res. Part A, Policy Pract.*, vol. 173, Jul. 2023, Art. no. 103686, doi: [10.1016/j.tra.2023.103686](https://doi.org/10.1016/j.tra.2023.103686).
- [101] Y. He, J. Guo, and X. Zheng, "From surveillance to digital twin: Challenges and recent advances of signal processing for industrial Internet of Things," *IEEE Signal Process. Mag.*, vol. 35, no. 5, pp. 120–129, Sep. 2018.
- [102] R. B. Roy et al., "Digital twin: Current scenario and a case study on a manufacturing process," *Int. J. Adv. Manuf. Technol.*, vol. 107, pp. 3691–3714, Apr. 2020.
- [103] S. Campbell et al., "Sensor technology in autonomous vehicles: A review," in *Proc. 29th Irish Signals Syst. Conf. (ISSC)*, 2018, pp. 1–4.
- [104] Q. Deng, X. Li, P. Ni, H. Li, and Z. Zheng, "Enet-CRF-Lidar: Lidar and camera fusion for multi-scale object recognition," *IEEE Access*, vol. 7, pp. 174335–174344, 2019, doi: [10.1109/ACCESS.2019.2956492](https://doi.org/10.1109/ACCESS.2019.2956492).
- [105] D. M. Botín-Sanabria, S. Mihaita, R. E. Peimbert-García, M. A. Ramírez-Moreno, R. A. Ramírez-Mendoza, and J. de J. Lozoya-Santos, "Digital twin technology challenges and applications: A comprehensive review," *Remote Sens.*, vol. 14, no. 6, p. 1335, Mar. 2022, doi: [10.3390/rs14061335](https://doi.org/10.3390/rs14061335).
- [106] Y. Lu, E. Shevtshenko, and Y. Wang, "Physics-based compressive sensing to enable digital twins of additive manufacturing processes," *J. Comput. Inf. Sci. Eng.*, vol. 21, no. 3, 2021, Art. no. 031009.
- [107] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, Apr. 2006.
- [108] S. Pop, L. Cristea, M. C. Luculescu, S. C. Zamfira, and A. L. Boer, "Vegetation index estimation in precision farming using custom multispectral camera mounted on unmanned aerial vehicle," in *Proc. 16th Int. Conf. Remote Eng. Virtual Instrum. Cyber-Phys. Syst. Digit. Twins*, 2020, pp. 674–685.
- [109] P. Furtner, E. Forstner, and A. Karlusch, "Automated infrastructure inspection based on digital twins and machine learning," in *Bridge Maintenance, Safety, Management, Life-Cycle Sustainability and Innovations*. Boca Raton, FL, USA: CRC Press, 2021, pp. 251–255.
- [110] A. Shabani, M. Skamantzari, S. Tapinaki, A. Georgopoulos, V. Plevris, and M. Kioumarsis, "3D simulation models for developing digital twins of heritage structures: challenges and strategies," *Procedia Struct. Integr.*, vol. 37, pp. 314–320, Jan. 2022.

- [111] M. M. Rathore, S. A. Shah, D. Shukla, E. Bentafat, and S. Bakiras, "The role of AI, machine learning, and big data in digital twinning: A systematic literature review, challenges, and opportunities," *IEEE Access*, vol. 9, pp. 32030–32052, 2021, doi: [10.1109/ACCESS.2021.3060863](https://doi.org/10.1109/ACCESS.2021.3060863).
- [112] Z. Ji, G. Shen, J. Wang, M. Collotta, Z. Liu, and X. Kong, "Multi-vehicle trajectory tracking towards digital twin intersections for Internet of Vehicles," *Electronics*, vol. 12, no. 2, p. 275, Jan. 2023, doi: [10.3390/electronics12020275](https://doi.org/10.3390/electronics12020275).
- [113] X. Xu, Z. Liu, M. Bilal, S. Vimal, and H. Song, "Computation offloading and service caching for intelligent transportation systems with digital twin," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 20757–20772, Nov. 2022, doi: [10.1109/TITS.2022.3190669](https://doi.org/10.1109/TITS.2022.3190669).
- [114] Y. Gao, S. Qian, Z. Li, P. Wang, F. Wang, and Q. He, "Digital twin and its application in transportation infrastructure," in *Proc. IEEE 1st Int. Conf. Digit. Twins Parallel Intell. (DTPI)*, 2021, pp. 298–301, doi: [10.1109/DTPI52967.2021.9540108](https://doi.org/10.1109/DTPI52967.2021.9540108).
- [115] J. Yang, R. S. Langley, and L. Andrade, "Digital twins for design in the presence of uncertainties," *Mech. Syst. Signal Process.*, vol. 179, Nov. 2022, Art. no. 109338, doi: [10.1016/j.ymssp.2022.109338](https://doi.org/10.1016/j.ymssp.2022.109338).
- [116] P. Aivaliotis, K. Georgoulis, and K. Alexopoulos, "Using digital twin for maintenance applications in manufacturing: State of the art and gap analysis," in *Proc. IEEE Int. Conf. Eng., Technol. Innov. (ICE/ITMC)*, 2019, pp. 1–5, doi: [10.1109/ICE.2019.8792613](https://doi.org/10.1109/ICE.2019.8792613).
- [117] J. Khalil, D. Yan, G. Guo, M. T. Sami, J. B. Roy, and V. P. Sisiopiku, "Traffic study of shared micromobility services by transportation simulation," in *Proc. IEEE Int. Conf. Big Data, Big Data*, 2021, pp. 3691–3699, doi: [10.1109/BigData52589.2021.9671455](https://doi.org/10.1109/BigData52589.2021.9671455).
- [118] P. A. Lopez et al., "Microscopic traffic simulation using SUMO," in *Proc. 21st Int. Conf. Intell. Transp. Syst. (ITSC)*, 2018, pp. 2575–2582, doi: [10.0/Linux-x86_64](https://doi.org/10.0/Linux-x86_64).
- [119] A. Osello, M. Fanshu, and F. M. Ugliotti, "Digital Twin at Urban Scale for a Master Plan in the Area of the CEBU Airport (Philippine) Supervisors Candidate," M.S. thesis, Automot. Eng., Politecnico di Torino, Turin, Italy, 2021.
- [120] Z. Xu, T. Jiang, and N. Zheng, "Developing and analyzing eco-driving strategies for on-road emission reduction in urban transport systems—A VR-enabled digital-twin approach," *Chemosphere*, vol. 305, Oct. 2022, Art. no. 135372, doi: [10.1016/j.chemosphere.2022.135372](https://doi.org/10.1016/j.chemosphere.2022.135372).
- [121] S. Khan, M. Farnsworth, R. McWilliam, and J. Erkoyuncu, "On the requirements of digital twin-driven autonomous maintenance," *Annu. Rev. Control*, vol. 50, pp. 13–28, Jan. 2020, doi: [10.1016/j.arcontrol.2020.08.003](https://doi.org/10.1016/j.arcontrol.2020.08.003).
- [122] D. Inoue, A. Okada, T. Matsumori, K. Aihara, and H. Yoshida, "Traffic signal optimization on a square lattice with quantum annealing," *Sci. Rep.*, vol. 11, no. 1, p. 3303, Dec. 2021, doi: [10.1038/s41598-021-82740-0](https://doi.org/10.1038/s41598-021-82740-0).
- [123] F. Neukart, G. Compostella, C. Seidel, D. Von Dollen, S. Yarkoni, and B. Parney, "Traffic flow optimization using a quantum annealer," *Frontiers ICT*, vol. 4, p. 29, Dec. 2017, doi: [10.3389/fict.2017.00029](https://doi.org/10.3389/fict.2017.00029).
- [124] S. Yarkoni et al., "Quantum shuttle: Traffic navigation with quantum computing," in *Proc. 1st ACM SIGSOFT Int. Workshop Archit. Paradig. Eng. Quantum Softw., Co-located (APEQS ESEC/FSE)*, 2020, pp. 22–30, doi: [10.1145/3412451.3428500](https://doi.org/10.1145/3412451.3428500).
- [125] M. Mashaly, "Connecting the twins: A review on digital twin technology & its networking requirements," *Procedia Comput. Sci.*, vol. 184, pp. 299–305, Jan. 2021, doi: [10.1016/j.procs.2021.03.039](https://doi.org/10.1016/j.procs.2021.03.039).
- [126] C. Campolo, G. Genovese, A. Molinaro, and B. Pizzimenti, "Digital twins at the edge to track mobility for MaaS applications," in *Proc. 2020 IEEE/ACM 24th Int. Symp. Distrib. Simul., Real Time Appl. (DS-RT)*, 2020, pp. 1–6, doi: [10.1109/DS-RT50469.2020.9213699](https://doi.org/10.1109/DS-RT50469.2020.9213699).
- [127] M. S. Salek et al., "A review on cybersecurity of cloud computing for supporting connected vehicle applications," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8250–8268, Jun. 2022, doi: [10.1109/JIOT.2022.3152477](https://doi.org/10.1109/JIOT.2022.3152477).
- [128] M. Hearn and S. Rix, "Cybersecurity considerations for digital twin implementations," *IIC J. Innov.*, vol. 10, pp. 107–113, Nov. 2019.
- [129] A. Sarker et al., "A review of sensing and communication, human factors, and controller aspects for information-aware connected and automated vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 1, pp. 7–29, Jan. 2020, doi: [10.1109/TITS.2019.2892399](https://doi.org/10.1109/TITS.2019.2892399).
- [130] S. E. Huang, W. Wong, Y. Feng, Q. A. Chen, Z. M. Mao, and H. X. Liu, "Impact evaluation of falsified data attacks on connected vehicle based traffic signal control," 2020, *arXiv:2010.04753*.
- [131] C. Xu, H. Wu, H. Liu, W. Gu, Y. Li, and D. Cao, "Blockchain-oriented privacy protection of sensitive data in the Internet of Vehicles," *IEEE Trans. Intell. Veh.*, vol. 8, no. 2, pp. 1057–1067, Feb. 2023, doi: [10.1109/TIV.2022.3164657](https://doi.org/10.1109/TIV.2022.3164657).
- [132] M. Dietz, B. Putz, and G. Pernul, "A distributed ledger approach to digital twin secure data sharing," in *Proc. 33rd Annu. Data Appl. Security Privacy*, 2019, pp. 281–300.
- [133] S. Suhail et al., "Blockchain-based digital twins: Research trends, issues, and future challenges," *ACM Comput. Surveys*, vol. 54, no. 11s, pp. 1–34, Sep. 2022, doi: [10.1145/3517189](https://doi.org/10.1145/3517189).
- [134] P. Lang, D. Tian, X. Duan, J. Zhou, Z. Sheng, and V. C. M. Leung, "Cooperative computation offloading in blockchain-based vehicular edge computing networks," *IEEE Trans. Intell. Veh.*, vol. 7, no. 3, pp. 783–798, Sep. 2022, doi: [10.1109/TIV.2022.3190308](https://doi.org/10.1109/TIV.2022.3190308).
- [135] K. Zhang, J. Cao, and Y. Zhang, "Adaptive digital twin and multiagent deep reinforcement learning for vehicular edge computing and networks," *IEEE Trans. Ind. Informat.*, vol. 18, no. 2, pp. 1405–1413, Feb. 2022, doi: [10.1109/TII.2021.3088407](https://doi.org/10.1109/TII.2021.3088407).
- [136] Z. Zhang and Q. Zhuang, "Distributed quantum sensing," *Quantum Sci. Technol.*, vol. 6, no. 4, Oct. 2021, Art. no. 043001, doi: [10.1088/2058-9565/abd4c3](https://doi.org/10.1088/2058-9565/abd4c3).
- [137] K. Bongs et al., "Taking atom interferometric quantum sensors from the laboratory to real-world applications," *Nat. Rev. Phys.*, vol. 1, no. 12, pp. 731–739, 2019.
- [138] L. Cohen, E. S. Matekole, Y. Sher, D. Istrati, H. S. Eisenberg, and J. P. Dowling, "Thresholded quantum LIDAR: Exploiting photon-number-resolving detection," *Phys. Rev. Lett.*, vol. 123, no. 20, 2019, Art. no. 203601.
- [139] Q. Zhuang, Z. Zhang, and J. H. Shapiro, "Entanglement-enhanced lidars for simultaneous range and velocity measurements," *Phys. Rev. A*, vol. 96, no. 4, 2017, Art. no. 040304.
- [140] M. R. Grace, C. N. Gagatsos, Q. Zhuang, and S. Guha, "Quantum-enhanced fiber-optic gyroscopes using quadrature squeezing and continuous-variable entanglement," *Phys. Rev. Appl.*, vol. 14, no. 3, Sep. 2020, Art. no. 034065, doi: [10.1103/PhysRevApplied.14.034065](https://doi.org/10.1103/PhysRevApplied.14.034065).
- [141] P. Almasan et al., "Digital twin network: Opportunities and challenges," 2022, *arXiv:2201.01144*.
- [142] B. Kunzer, M. Berges, and A. Dubrawski, "The digital twin landscape at the crossroads of predictive maintenance, machine learning and physics based modeling," 2022, *arXiv:2206.10462*.
- [143] X. Wang, F. Liu, and D. Zhao, "Deep transfer fault diagnosis using digital twin and generative adversarial network," in *Proc. IEEE Int. Conf. Sens., Diagn., Progn., Control (SDPC)*, 2021, pp. 186–193.
- [144] S. P. Ramu et al., "Federated learning enabled digital twins for smart cities: Concepts, recent advances, and future directions," *Sustain. Cities Soc.*, vol. 79, Apr. 2022, Art. no. 103663, doi: [10.1016/j.scs.2021.103663](https://doi.org/10.1016/j.scs.2021.103663).
- [145] H. Hussain, M. B. Javaid, F. S. Khan, A. Dalal, and A. Khalique, "Optimal control of traffic signals using quantum annealing," *Quantum Inf. Process.*, vol. 19, no. 9, p. 312, 2020, doi: [10.1007/s11128-020-02815-1](https://doi.org/10.1007/s11128-020-02815-1).
- [146] S. S. Gill et al., "Quantum computing: A taxonomy, systematic review and future directions," *Softw. Pract. Exp.*, vol. 52, no. 1, pp. 66–114, 2022.
- [147] S. A. Niederer, M. S. Sacks, M. Girolami, and K. Willcox, "Scaling digital twins from the artisanal to the industrial," *Nat. Comput. Sci.*, vol. 1, no. 5, pp. 313–320, 2021, doi: [10.1038/s43588-021-00072-5](https://doi.org/10.1038/s43588-021-00072-5).



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